

FERMION CARRIER-GENERATION STRUCTURE III

Structural Universality and Exact Terminal Bifurcation

One-Higgs naturality, projection firewalls,
and the branch-complete fermionic endpoint

The exact three-root structural architecture is universal across the neutral, charged-lepton, up-coloured, and down-coloured matter kinds. Terminal physical standing is not. This paper proves the lawful cross-kind integration theorem: one common structural reduct, kind-preserving fibres, maximal-domain naturality, one Higgs bridge/fixation carrier, a typed partial projection and the process fibration, an exact evidence-relative terminal status atlas, and a fixed-authority global no-backflow theorem. Neutral one-bearer closure is retained only on its two certified child scopes; the neutral parent complement and the charged and coloured terminal branches retain their exact readiness obstructions. The result is universal where the proof is universal and bifurcated where terminal authority is not yet common.

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Primitive-standing reference edition | 21 September 2026

Abstract

The first two papers of the Fermion Carrier–Generation Structure programme separate two questions usually displayed on one Standard Model table. FCGS-I forms an unlabelled three-root object I_3 , the typed matter bundle $I_3 \times \{N, L, U, D\}$, and three rooted structural Higgs-response incidences per matter kind through a single bridge/fixation carrier. FCGS-II defines target-relative semantic carrier identity and terminal realization, proves complete-signature and assignment-orbit factorisation on the activated interfaces, and freezes exact readiness-valued terminal dispositions for the charged and coloured kinds. The present paper integrates those accepted handoffs without modifying either theorem chain.

We construct the fermion-kind category as a tagged family of lawful same-target domains. Full-target cross-kind arrows are absent unless independently licensed; after explicit erasure to the common structural reduct, the kind-indexed atlas admits effective codescent. On each maximal lawful domain, the accepted correspondences

$$\text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} Q_f^Y = \text{Disc}(A_f), \quad E_f = I_3 \times \{r_f\}, \quad f \in \{N, L, U, D\},$$

are natural equivalences, and every accepted rooted response factors through the same $C_H = H_{\text{br}/\text{fix}}$. This proves universal structural exact-three closure without identifying matter kinds, importing particle names, or turning a Yukawa display into a cardinality argument.

Terminal standing is represented by a tagged disposition family rather than an analogy. The neutral fibre contains two certified child scopes with one terminal bearer and a distinct three-light-plus-one-heavy spectral decomposition; its unresolved parent complement remains typed. The charged-lepton fibre has disposition $\text{NotReady}[(\text{CL}, \mathbf{M}_L^0)]$, and the two coloured fibres have $\text{NotReady}[(\text{Confinement}, \mathbf{M}_U^0)]$ and $\text{NotReady}[(\text{Confinement}, \mathbf{M}_D^0)]$. Their terminal ROC objects are unformed and are neither counted nor completed by cross-kind analogy. Consequently the certified universal-terminal activation set and the certified common-apex witness set are empty at the frozen evidence bases, although no physical nonexistence theorem is asserted.

Carrier-preserving projection, the particle dictionary, and the process fibration are therefore explicitly partial. Masses, matrices, singular values, poles, fits, detector records, and the HNL scouting coordinates remain downstream data and cannot reverse-authorize a carrier, a root count, or a common apex. The capstone is an exact branch-complete endpoint: **StructuralUniversal** at premetric root-response scope, **TerminalPartial** at terminal physical scope, and **NotReady** exactly where required certificates or common-target fields are absent.

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1 Import finality and the frozen successor target

FCGS–III is a successor theorem whose premises include the closed FCGS–I result and the sealed FCGS–II handoff. We therefore begin by fixing exactly what is imported, exactly which parts of an import are theorem-bearing, and exactly which parts are only status-bearing. This separation is essential: a status value can block a successor map, but it cannot manufacture the mathematical object whose absence it records.

The term *premetric* is used in the AASC sense: admissible structural standing is fixed before a metric description is licensed [5]. The requirement that a realized physical theory admit a gravitationally coherent metric description is relevant background [8], but it is not a constructor or dynamical-gravity premise in FCGS–III.

1.1 The two accepted predecessor results

Let

$$\mathfrak{F} = \{N, L, U, D\}$$

be the four fermion kinds. The canonical structural antecedent is the accepted FCGS–I primitive-standing reference edition theorem [1], denoted by $\mathbb{H}_I^{\text{prim}}$. For every $f \in \mathfrak{F}$, it establishes the fixed three-point root object

$$I_3 := \text{Occ}_N, \quad |I_3| = 3, \quad \text{Aut}(K_3) \cong S_3,$$

the rooted structural object $\mathcal{S}_f(C_f)$, the response-slot set A_f , and the accepted typed chain

$$\text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{x_f} Q_f^Y = \text{Disc}(A_f). \quad (1.1)$$

Both arrows in (1.1) are equivalences in the structural groupoid. The occurrence sets for L, U, D are separately registered inputs; the common root is not an independent recount of those occurrence sets.

Here the positioned fibre and its canonical projection are

$$E_f := I_3 \times \{r_f\}, \quad p_f : E_f \xrightarrow{\sim} I_3.$$

Only after invoking this fixed-kind identification may the chain be displayed with source $\text{Disc}(I_3)$. Its root-normalised first arrow is the separately induced composite

$$\eta_f^{I_3} := \eta_f \circ \text{Disc}(p_f^{-1}) : \text{Disc}(I_3) \longrightarrow \mathcal{S}_f(C_f). \quad (1.2)$$

Equation (1.2) does not rename or redefine η_f , whose canonical source remains $\text{Disc}(E_f)$. Nor does the fixed-kind identification supply a cross-kind full-target arrow.

The retained count warrant. The antecedent includes the original P_0 work, its fully occupied primitive architecture I , and the source-root identification $\sigma_N : I \xrightarrow{\sim} I_3$ constructed in FCGS–I, Sections 3–4. Adequacy means one joint construction of all original duties; primitive standing means adequacy with no adequate proper subarchitecture. These are count-open conditions. The upstream result $\text{Prim}_{P_0}(I) \iff |I| = 3$ [9, Theorem 6.2] therefore counts the received root object. Its rank-one circuit register follows after the count [9, Corollary 5.5]. Every accepted structural transport below preserves this original-work receipt and complete occupation. Additional terminal, response, or spectral coordinates retain their own types and give no additional primitive generation standing for the unchanged work [9, Theorem 6.4]. Reader descent beyond the original interface requires the actual fibre test [9, Proposition 6.7]; equality of cardinality alone is insufficient.

The accepted terminal antecedent is the FCGS–II primitive-standing reference edition theorem [2], denoted by $\mathbb{H}_I^{\text{prim}}$. It contains the evaluated N01–N18 records, including the following two logically different kinds of data.

- (i) N01 and N13 seal the immutability of the structural chain and its positioned-fibre diagrams. The $\text{Disc}(I_3)$ -source shorthand appearing in N01 and N13 is read against the authoritative FCGS–I primitive-standing reference edition signature: the canonical arrow is $\eta_f : \text{Disc}(E_f) \rightarrow \mathcal{S}_f(C_f)$, while the $\text{Disc}(I_3)$ -source arrow is $\eta_f^{I_3}$ from (1.2). This type normalisation changes no FCGS–II gate disposition.
- (ii) N18 is a successor-stage constraint: no uncertified positive terminal packet for the evaluated L, U, D targets is admitted into the successor object. In particular, an empty admitted positive summand is not a theorem that a positive occupant does not exist; it is the consequence of a failed admission certificate at the stated target.

The neutral ready-child records, where present, remain child-scope records. They are not promoted to a common four-kind terminal apex by being placed next to the N18 ledger.

Definition 1.1 (Fixed successor input). *The fixed successor input at the declared evidence basis is the typed pair*

$$\mathbb{H}_{III}^{\text{in}} := (\mathbb{H}_I^{\text{prim}}, \mathbb{H}_{II}^{\text{prim}}),$$

with the following formation rule. Structural objects and arrows may be formed only from the theorem-bearing component $\mathbb{H}_I^{\text{prim}}$. Terminal records may be carried forward only as tagged child-scope objects or as tagged dispositions. A positive terminal object may enter a successor construction only through the N18 admission map, which is defined precisely on certified positive terminal packets.

Thus the N18 input inclusion has the schematic type

$$j_{18} : \coprod_{f \in \mathfrak{F}} \text{ActCert}_f^+ \longrightarrow \mathbb{H}_{III}^{\text{in}},$$

where ActCert_f^+ is the groupoid of *actually certified positive* terminal packets at the evaluated target. For the evaluated charged and quark branches, the FCGS–II handoff supplies no object of this groupoid. There is consequently no term on which j_{18} could act for those branches.

1.2 Import ledger

The following ledger is an index to the imports. It is not a substitute for the definitions and proofs below.

FCGS–I primitive-standing reference edition.

The import is I_3 , the positioned fibres, rooted objects, response slots, and (1.1). Its lawful uses are structural exact-three reasoning, same-target transport, and explicitly erased structural codescent. It supplies no new L, U, D count, positive Yukawa or mass values, or terminal common apex.

FCGS–I primitive-standing reference edition naturality theorem.

The import is naturality of η_f and χ_f on the complete-target-preserving domain. It forms the maximal lawful same-target groupoids of Section 2; it supplies neither a cross-kind full-target arrow nor naturality on an uncertified map.

FCGS–I primitive-standing reference edition codescent theorem.

The import is effective codescent of the four structural reducts over $\mathcal{A}_0$. It licenses comparison only after the explicit erasures in Definition 3.3; it does not identify the four full targets or erase terminal dispositions.

FCGS–II primitive-standing reference edition, N01 and N13.

The import is structural immutability together with the shorthand appearing in N01 and N13, resolved against the canonical positioned-fibre signature. Fixed-kind diagrams may be type-checked through p_f , but shorthand may not replace the canonical source $\text{Disc}(E_f)$ of η_f .

FCGS–II primitive-standing reference edition, N18.

The import is certified-packet-only successor admission and the empty evaluated L, U, D positive summand. It preserves **NotReady** and the absence of an admitted terminal object; it supplies no no-go theorem, negative occupant, or imputed terminal value.

Combined v2.0.0 plan, U01–U04.

The plan fixes scope, ordering, and review obligations. It governs the argument but supplies no mathematical premise or terminal conclusion.

The proof classes are therefore different. The FCGS–I rows import proved structural equivalences and effective descent. The FCGS–II rows import a sealed typed disposition and an admission rule. The plan row imports no scientific fact at all; it merely fixes the comparison protocol.

Theorem 1.2 (U01 import finality). *Every well-typed FCGS–III structural term factors through the FCGS–I primitive-standing reference edition structural antecedent, and every FCGS–III terminal successor term factors through the N18 certified-packet admission domain. Consequently:*

- (a) *the chain (1.1) is unchanged;*
- (b) *$\eta_f^{I_3}$ is induced as in (1.2) and does not replace η_f ;*
- (c) *no evaluated L, U, D positive terminal object is present in the successor input; and*
- (d) *no child-scope neutral terminal record becomes a universal terminal record merely by import.*

Proof. By Definition 1.1, the structural formation rules have only one theorem-bearing structural source, namely $\mathbb{H}_f^{\text{prim}}$. Their factorisation through that packet is therefore part of the type of a lawful FCGS–III structural term. In particular, the named arrows are the released η_f and χ_f , so shorthand in the terminal ledger cannot give the displayed composite a new source or target. The root-normalised map is instead obtained by precomposition with $\text{Disc}(p_f^{-1})$, proving (a) and (b).

For terminal data, N18 defines the successor admission domain to be ActCert_f^+ . The sealed charged and quark records contain no certified positive object, hence contribute no object to the coproduct on which j_{18} is defined. This proves (c). Finally, the neutral records retain their child-scope target tags. Forgetting such a tag is not among the frozen formation rules, and no comparison theorem identifies those targets with all four fermion targets. Hence no common terminal record follows, proving (d). $\square$

Lemma 1.3 (Status values do not supply missing witnesses). *For a terminal target T , a ledger value $\text{NotReady}(T)$ cannot be used as an object of $\text{ActCert}^+(T)$, as a morphism into T , or as a proof of the negation of such an object.*

Proof. These expressions have different types. The first is a disposition in the status ledger. The second would be a positive certificate carrying a target, a bearer, and its verified registry rows. The third would require a negative mathematical theorem. No coercion among these types is present in either frozen handoff. Therefore none is available in the successor theory. $\square$

Counterexample criterion 1.4 (Import-finality falsifiers). *The U01 layer fails if any downstream section does one of the following: uses an earlier noncanonical FCGS–I signature as the theorem source; changes the source of η_f ; lets root-normalised shorthand in the terminal ledger override the canonical*

positioned-fibre signature; treats $\eta_f^{I_3}$ as a renamed η_f ; drops an N18 NotReady tag; inserts an uncertified charged or quark terminal packet; uses a plan table as proof of a mathematical premise; or promotes a child-scope neutral record to a four-kind terminal apex without a new typed comparison theorem.

Status theorem 1.5 (Claim boundary after U01). *The successor input is final at the import boundary. It supports the closed structural chain and preserves the sealed terminal dispositions. It does not yet assert cross-kind comparison, codescent, a universal terminal bearer, or a common physical apex.*

2 The fermion-kind category and lawful comparison modes

The four fermion kinds share a certified structural root, but they do not share a complete physical target. A rigorous universality statement must therefore distinguish three comparison modes: morphisms of full targets, spans over the common root, and morphisms obtained only after an explicit structural erasure. This section constructs the first two. The third is constructed in Section 3.

2.1 Full target fingerprints

For an FCGS–I presentation X of kind f , write

$$\Phi_f(X) := (f, \text{source rows, occurrence rows, root rows, response rows, registry rows,} \\ \text{charge, colour, weak component, chirality, Higgs incidence, confinement,} \\ \text{status, boundary and continuation rows, provenance, failure data})$$

for its *complete target fingerprint*. The displayed list is not a collection of optional metadata. It specifies which coordinates a full target morphism must preserve. In particular, the kind tag is part of the target type.

Definition 2.1 (Maximal lawful same-target groupoid). *For $f \in \mathfrak{F}$, let $\mathcal{K}_f$ be the category whose objects are certified presentations X with complete target fingerprint $\Phi_f(X)$. A morphism $u : X \rightarrow Y$ is an invertible transport of all fields of Φ_f such that*

- (i) *it commutes with the root projection, rooted quotient, response-slot map, and all registry structure maps;*
- (ii) *it preserves the target, status, boundary, continuation, provenance, and failure rows extensionally; and*
- (iii) *its inverse satisfies the same two conditions.*

No additional restriction is imposed. Thus $\mathcal{K}_f$ contains every lawful invertible same-target comparison and is maximal with respect to the complete fingerprint.

Lemma 2.2 (Same-target closure). *Each $\mathcal{K}_f$ is a groupoid. Moreover, any invertible comparison of two kind- f presentations that preserves the complete target fingerprint is already a morphism of $\mathcal{K}_f$.*

Proof. The identity transport commutes with every structure map and fixes every fingerprint row. If $u : X \rightarrow Y$ and $v : Y \rightarrow Z$ have these properties, then $v \circ u$ does as well: commutative squares compose, and equality of each preserved row is transitive. Closure under inverse is required in the definition. Hence $\mathcal{K}_f$ is a groupoid. The final assertion is immediate from the absence of any condition beyond invertibility and complete-fingerprint preservation; it is the promised maximality, rather than an assertion that a smaller convenient subcategory is sufficient. $\square$

Definition 2.3 (Full fermion-target category). *The full fermion-target category is the disjoint coproduct*

$$\mathbf{K}_{\mathfrak{F}}^{\text{full}} := \coprod_{f \in \mathfrak{F}} \mathcal{K}_f.$$

Equivalently, its Hom sets are

$$\text{Hom}_{\mathbf{K}_{\mathfrak{F}}^{\text{full}}}(X_f, Y_g) = \begin{cases} \text{Hom}_{\mathcal{K}_f}(X_f, Y_f), & f = g, \\ \emptyset, & f \neq g. \end{cases} \quad (2.1)$$

The second line of (2.1) is a scientific type boundary, not a claim that the four sectors have no common structure. A full-target arrow must preserve the first coordinate of Φ_f . If it carried an object of kind f to one of kind g , preservation would imply $f = g$. For distinct tags this is impossible. Shared root cardinality does not alter that argument because cardinality is only one coordinate of a much larger fingerprint.

2.2 Root-relative comparison

For every $f \in \mathfrak{F}$, the canonical packet supplies

$$E_f = I_3 \times \{r_f\}, \quad p_f : E_f \xrightarrow{\sim} I_3.$$

Thus two distinct kinds admit the span

$$\text{Disc}(E_f) \xrightarrow{\text{Disc}(p_f)} \text{Disc}(I_3) \xleftarrow{\text{Disc}(p_g)} \text{Disc}(E_g). \quad (2.2)$$

This is a comparison over a common structural base. It is not a morphism in $\mathbf{K}_{\mathfrak{F}}^{\text{full}}$. Even though $p_g^{-1} \circ p_f : E_f \rightarrow E_g$ is a bijection of positioned fibres, it has forgotten precisely the target coordinates that separate kind f from kind g . It can therefore transport root positions, but not charge, colour, chirality, confinement, kind-specific Higgs incidence, terminal status, or the complete provenance record.

Definition 2.4 (Typed comparison modes). *For $f, g \in \mathfrak{F}$, an FCGS-III comparison is declared to be exactly one of the following typed objects.*

- (a) A full-target comparison: a morphism in $\mathbf{K}_{\mathfrak{F}}^{\text{full}}$, necessarily with $f = g$.
- (b) A root-relative comparison: a span of the form (2.2), retaining its two kind tags.
- (c) A reduct comparison: a morphism formed after the explicit erasure functors of Definition 3.3 have landed in the common reduct $\mathcal{A}_0$.

A name match, a common cardinality, or a shared symbol is not a fourth comparison mode.

Theorem 2.5 (U02 exact typed noncomparability). *For every $f, g \in \mathfrak{F}$:*

- (i) all lawful invertible same-target comparisons occur in $\mathcal{K}_f$;
- (ii) if $f \neq g$, then $\text{Hom}_{\mathbf{K}_{\mathfrak{F}}^{\text{full}}}(X_f, Y_g) = \emptyset$;
- (iii) the common-root span (2.2) is lawful for all pairs (f, g) , but it does not induce a full-target arrow; and
- (iv) in particular, no full-target U -to- D equivalence is present.

Proof. Part (i) is Lemma 2.2. For (ii), suppose a full-target arrow $u : X_f \rightarrow Y_g$ existed. By definition it preserves every coordinate of the complete fingerprint, including the kind coordinate. Hence $f = g$, contradicting the hypothesis. This proves the empty Hom-set formula without appealing to a naming convention.

For (iii), both legs of (2.2) are imported equivalences of discrete root groupoids, so the span is well typed. However, neither leg has a full fermion target as its codomain. Composing through the root therefore gives only a root-position bijection. The erased target coordinates cannot be reconstructed from that bijection, so no morphism of $\mathbf{K}_{\mathfrak{F}}^{\text{full}}$ follows. Part (iv) is the special case $(f, g) = (U, D)$. $\square$

Corollary 2.6 (The accepted chain remains fibrewise). *For each f , the chain*

$$\text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} Q_f^Y = \text{Disc}(A_f)$$

belongs to the kind- f structural fibre. Only after invoking $p_f : E_f \xrightarrow{\sim} I_3$ may its first leg be displayed with source $\text{Disc}(I_3)$, in which case the induced map is $\eta_f^{I_3} = \eta_f \circ \text{Disc}(p_f^{-1})$. No composite of these expressions has type $X_f \rightarrow Y_g$ in the full-target category when $f \neq g$.

Proof. The first two assertions use the canonical FCGS-I primitive-standing reference edition signature, with the FCGS-II N01/N13 immutability record read through the fixed-kind root-normalisation of Section 1. The last follows from (2.1); an expression with a cross-kind full-target type would require an inhabitant of an empty Hom set. $\square$

Counterexample criterion 2.7 (Fermion-category falsifiers). *The U02 construction is falsified by any purported full-target arrow whose source and target carry distinct kind tags; by deriving such an arrow merely from $|I_3| = 3$; by calling $p_D^{-1}p_U$ a U -to- D target equivalence; by deleting charge, colour, chirality, confinement, status, or provenance before declaring that a comparison is “full”; or by treating an empty cross-kind Hom set as a proof that no common reduct can exist.*

Status theorem 2.8 (Claim boundary after U02). *The largest lawful full-target comparison domain is the disjoint union of the four same-target groupoids. Cross-kind root-relative spans are available, but full-target cross-kind Hom sets are empty. This is explicit noncomparability at full target, not terminal universality and not a physical no-go theorem.*

3 Structural exact-three universality on the lawful domain

We now state the strongest universality result supported by the frozen handoffs. It has two parts. First, the accepted chain is natural on every lawful same-target arrow, and on no larger full-target domain claimed here. Second, the four kind charts admit effective codescent only after each chart has been sent to the same explicitly specified structural reduct. Neither part asserts a terminal object in the full fermion-target category.

3.1 The maximal naturality domain

Fix $f \in \mathfrak{F}$. On $\mathcal{K}_f$, define three groupoid-valued functors by

$$\mathbb{E}_f(X) := \text{Disc}(E_f(X)), \quad \mathbb{S}_f(X) := \mathcal{S}_f(C_f; X), \quad \mathbb{A}_f(X) := \text{Disc}(A_f(X)).$$

If $u : X \rightarrow Y$ is a morphism of $\mathcal{K}_f$, let $\mathbb{E}_f(u)$ be its certified positioned-fibre transport, let $\mathbb{S}_f(u)$ be the induced transport of rooted components, and let $\mathbb{A}_f(u)$ be its induced transport of response slots. These maps exist because a morphism of $\mathcal{K}_f$ preserves the root projection, rooted quotient, response map, and complete registry rows.

Definition 3.1 (Fibrewise structural transformations). *For each kind f , the components of the structural transformations are the released arrows*

$$\eta_{f,X} : \mathbb{E}_f(X) \longrightarrow \mathbb{S}_f(X), \quad \chi_{f,X} : \mathbb{S}_f(X) \longrightarrow \mathbb{A}_f(X).$$

Only after the canonical fixed-kind identification $p_{f,X} : E_f(X) \xrightarrow{\sim} I_3(X)$ is invoked may the first component be displayed with source $\text{Disc}(I_3(X))$. That induced map is

$$\eta_{f,X}^{I_3} = \eta_{f,X} \circ \text{Disc}(p_{f,X}^{-1}) : \text{Disc}(I_3(X)) \longrightarrow \mathbb{S}_f(X).$$

Theorem 3.2 (U03 maximal lawful same-target naturality). *For every $f \in \mathfrak{F}$,*

$$\eta_f : \mathbb{E}_f \Longrightarrow \mathbb{S}_f, \quad \chi_f : \mathbb{S}_f \Longrightarrow \mathbb{A}_f$$

are natural equivalences on $\mathcal{K}_f$. Explicitly, for every lawful same-target arrow $u : X \rightarrow Y$,

$$\mathbb{S}_f(u) \circ \eta_{f,X} = \eta_{f,Y} \circ \mathbb{E}_f(u), \tag{3.1}$$

$$\mathbb{A}_f(u) \circ \chi_{f,X} = \chi_{f,Y} \circ \mathbb{S}_f(u). \tag{3.2}$$

The domain is maximal among lawful complete-target-preserving same-kind arrows. No cross-kind full-target naturality assertion is made.

Proof. The component $\eta_{f,X}$ is obtained from the certified root projection and rooted-component equivalence, while $\chi_{f,X}$ is obtained from the certified rooted response quotient. A morphism of $\mathcal{K}_f$ commutes with each of those structure maps by Definition 2.1. Following an element first across a released component and then along the transport therefore gives the same root component or response slot as transporting first and applying the released component afterward. This is exactly (3.1) and (3.2). Since the released components are equivalences at every object, both transformations are natural equivalences.

Maximality follows from the construction of $\mathcal{K}_f$: every invertible same-kind arrow preserving the complete target fingerprint is already a morphism of that groupoid. Enlarging the domain with an arrow that fails one of those preservation requirements would no longer be a lawful full-target extension. Finally, by (2.1), there is no full-target arrow between different kind components on which an additional naturality square could even be typed. $\square$

The four transformations can be written compactly as a coproduct

$$\coprod_{f \in \mathfrak{F}} \eta_f \quad \text{and} \quad \coprod_{f \in \mathfrak{F}} \chi_f$$

on $\mathbf{K}_{\mathfrak{F}}^{\text{full}}$. This notation means “one natural transformation on each connected component.” It supplies no cross-kind component and imposes no cross-kind naturality square, because the relevant Hom sets are empty.

3.2 Explicit erasure and the common reduct

The common structural object is not obtained by pretending that the four full targets coincide. It is obtained by specifying a reduct and proving descent for the reduct charts.

We use the inherited FCGS-I names $D_{\text{gen}}^{\text{open}}$ for the full subdomain on which the generation-root and response-slot structures are formed, Ω_{prim} for the primitive rank-one circuit/holonomy residue object, ∂_0 for the common premetric boundary interface, and Cont_0 for its structural continuation relation [1]. These are coordinates of the accepted structural antecedent, not new assumptions of FCGS-III. In particular, rank one is the circuit rank of the already proved three-root architecture. It is not a separate premise used to obtain the root count [9, Corollary 5.5 and Theorem 6.2].

Definition 3.3 (The common structural erasure). *For each $f \in \mathfrak{F}$, let $\mathcal{A}_f^{\text{str}}$ be the kind-tagged structural chart of the accepted target. Define*

$$e_f : \mathcal{A}_f^{\text{str}} \longrightarrow \mathcal{A}_0$$

to retain exactly the coordinates

$$(D_{\text{gen}}^{\text{open}}, I_3, K_3, \Omega_{\text{prim}}, H_{\text{br/fix}}, \text{RootSchema}, \text{ResponseSlotSchema}, \partial_0, \text{Cont}_0), \quad (3.3)$$

and to erase exactly the kind tag, charge, colour, weak component, chirality, kind-specific Higgs incidence, confinement data, terminality data, spectra, and metric data. The erased fields are not silently set equal; they are absent from the codomain type $\mathcal{A}_0$.

The source category in Definition 3.3 has only structural morphisms; the target-specific fields are inert annotations rather than extra morphism degrees of freedom. Thus the canonical FCGS–I statement that each e_f is fully faithful and essentially surjective is compatible with the fact that the annotations are erased.

For $f, g \in \mathfrak{F}$, form the overlap only over the common reduct:

$$\mathcal{A}_{fg}^0 := \mathcal{A}_f^{\text{str}} \times_{\mathcal{A}_0} \mathcal{A}_g^{\text{str}}. \quad (3.4)$$

The imported transition equivalences on these overlaps obey the identity law, the inverse law, and the triple-overlap cocycle law. Write

$$\text{Codesc}_0(\{e_f\}_{f \in \mathfrak{F}})$$

for the groupoid codescent object of this erased Cech diagram. The subscript zero records that the diagram has already landed over $\mathcal{A}_0$; there is no codescent object of the four full physical targets in this notation. The atlas itself remains tagged:

$$\mathcal{A}_{\text{atlas}}^{\text{tag}} := \coprod_{f \in \mathfrak{F}} \{f\} \times \mathcal{A}_f^{\text{str}}.$$

Thus e_f forgets the tag only in its image in $\mathcal{A}_0$; it does not delete the chart index from the atlas or turn a reduct identification into a full-target cross-kind arrow.

Theorem 3.4 (U04 effective erased codescent). *The canonical comparison functor*

$$\kappa : \text{Codesc}_0(\{e_f\}_{f \in \mathfrak{F}}) \longrightarrow \mathcal{A}_0 \quad (3.5)$$

is an equivalence of groupoids. More precisely, it is fully faithful and essentially surjective. This equivalence is formed only after the erasures in Definition 3.3; it neither creates nor reflects a cross-kind arrow in $\mathbf{K}_{\mathfrak{F}}^{\text{full}}$.

Proof. The FCGS–I codescent theorem [1] supplies three ingredients: each chart functor e_f is fully faithful, each is essentially surjective onto the common reduct, and the overlap transitions satisfy the identity, inverse, and triple cocycle laws. Hence any morphism represented in one chart has a unique compatible representative on every overlap. Two local representatives have the same image under κ precisely when the overlap relation identifies them. This proves fullness and faithfulness of κ .

For essential surjectivity, let a be an object of $\mathcal{A}_0$. Essential surjectivity of any e_f gives a chart object a_f and an isomorphism $e_f(a_f) \cong a$. The cocycle identifies different choices on pairwise and triple overlaps, so a_f determines an object of the codescent groupoid whose image is isomorphic to a . Therefore κ is an equivalence.

This argument takes place entirely in the categories appearing in (3.4). The erased coordinates are not objects of the Cech diagram, and no inverse functor reconstructing them from $\mathcal{A}_0$ is part of the packet. Consequently the equivalence cannot be lifted to a cross-kind full-target arrow. The tagged atlas retains all four chart indices, and Equation (2.1) remains unchanged. $\square$

3.3 The fixed-domain exact-three theorem

Theorem 3.5 (FCGS–III structural universality). *At the frozen FCGS–I primitive-standing reference edition target, the following statements hold for every $f \in \mathfrak{F}$.*

(i) *The exact accepted chain is*

$$\text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} Q_f^Y = \text{Disc}(A_f),$$

and both arrows are equivalences.

(ii) *The chain is natural on the maximal lawful same-target groupoid $\mathcal{K}_f$.*

(iii) *Each of $\mathcal{S}_f(C_f)$ and $\text{Disc}(A_f)$ has exactly three connected structural components:*

$$|\pi_0 \mathcal{S}_f(C_f)| = |\pi_0 \text{Disc}(A_f)| = |I_3| = 3.$$

(iv) *The four explicitly erased structural charts have effective codescent*

$$\text{Codesc}_0(\{e_f\}) \simeq \mathcal{A}_0.$$

(v) *Full-target cross-kind Hom sets remain empty.*

These five clauses constitute StructuralUniversal in this paper. They do not assert that $\mathcal{A}_0$ is terminal among physical targets, that one bearer occupies all four kinds, or that any unresolved terminal branch is positive.

Proof. Clause (i) is the frozen chain of Theorem 1.2. Clause (ii) is Theorem 3.2. Since equivalences of groupoids induce bijections on connected components, clause (i) gives

$$\pi_0 \mathcal{S}_f(C_f) \cong \pi_0 \text{Disc}(E_f) \cong E_f$$

and likewise $\pi_0 \text{Disc}(A_f) \cong E_f$. The fixed-kind bijection $p_f : E_f \xrightarrow{\sim} I_3$, followed by the imported antecedent $|I_3| = 3$, proves clause (iii). Clause (iv) is Theorem 3.4, and clause (v) is Theorem 2.5. The final nonclaims follow from the types used in the proof: terminality, spectra, metric data, and kind-specific physical incidences are erased before clause (iv), while clauses (i)–(iii) concern structural groupoids only. No step produces a common terminal bearer or a universal physical apex. $\square$

Corollary 3.6 (Twelve tagged structural incidences). *The tagged disjoint atlas contains*

$$\sum_{f \in \mathfrak{F}} |\pi_0 \mathcal{S}_f(C_f)| = 4 \cdot 3 = 12$$

rooted structural incidences. This is a count of tagged structural slots through the one frozen bridge schema $H_{\text{br}/\text{fix}}$; it is not a count of positive Yukawa values, masses, terminal carriers, or physically identified cross-kind occupants.

Proof. There are four retained kind tags in the disjoint atlas and, by Theorem 3.5(iii), three structural components in each chart. Disjointness makes the cardinalities additive. The excluded interpretations use fields absent from the structural component count, so they do not follow from the arithmetic. $\square$

Counterexample criterion 3.7 (Structural-universality falsifiers). *The U03–U04 result fails if naturality is asserted for an uncertified or cross-kind full-target map; if codescent is formed before the explicit erasure to $\mathcal{A}_0$; if any field in (3.3) is dropped; if an erased charge, colour, chirality, confinement, terminal, spectral, or metric field is silently reconstructed; if the structural codescent equivalence is called a full-target equivalence; if a table replaces the elementwise naturality or codescent argument; or if the exact-three component count is reported as three positive masses or three terminal bearers.*

Status theorem 3.8 (Strongest U01–U04 endpoint). *The four imported fermion kinds are structurally exact-three and admit a single effective common reduct after explicit erasure. Naturality is maximal on lawful same-target arrows, while full-target cross-kind Hom sets are empty. This is a fixed-domain structural universality theorem. Terminal universality remains unproved and is not part of the endpoint.*

4 Exact terminal bifurcation

Structural universality does not make terminal physical standing universal. The distinction is already forced by the types. The structural theorem has source $\text{Disc}(E_f)$, where $E_f = I_3 \times \{r_f\}$, and codomain the rooted structural and response-slot objects

$$\text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{x_f} Q_f^Y = \text{Disc}(A_f). \quad (4.1)$$

A terminal theorem instead requires a target-relative bearer, a complete continuation signature, a formed terminal quotient, and a positive role-occupancy certificate. None of those objects is supplied merely by the three components of (4.1).

4.1 The terminal status fibre

Let

$$\text{TStat} := \{\text{Positive}[-], \text{Negative}[-], \text{Split}[-], \text{TargetDependent}[-], \text{NotReady}[-]\}$$

be the tagged codomain of terminal dispositions. The argument of a status is part of the value: two readiness values with different residual packets are not interchangeable.

For the neutral theorem, write

$$B_N^{\text{ready}} := \{b_{\text{acc}}, b_{\text{glob}}\}$$

for the two completed vectorlike-joint child scopes of PCR-6 [3]. Let PhysMod_b be the category of physical realization models satisfying the complete target contract of the ready child b . If $b \in B_N^{\text{ready}}$ and $X \in \text{PhysMod}_b$, let $\text{TermCar}_b(X)$ denote its terminal-bearer quotient and let $\text{SpecCore}_b(X)$ denote its standing spectral core.

Definition 4.1 (Frozen terminal disposition). *The frozen terminal disposition is the target-indexed partial evaluation $\text{Disp}^{\text{term}}$ with the following exact values:*

$$\text{Disp}_N^{\text{term}}(b; X) = \text{Positive}[|\text{TermCar}_b(X)| = 1], \quad b \in B_N^{\text{ready}}, \quad (4.2)$$

$$\text{Disp}_N^{\text{term}}(\text{parent} \setminus B_N^{\text{ready}}) = \text{NotReady}[\text{PCR5E}], \quad (4.3)$$

$$\text{Disp}_L^{\text{term}} = \text{NotReady}[(\text{CL}, \mathbf{M}_L^0)], \quad (4.4)$$

$$\text{Disp}_U^{\text{term}} = \text{NotReady}[(\text{Confinement}, \mathbf{M}_U^0)], \quad (4.5)$$

$$\text{Disp}_D^{\text{term}} = \text{NotReady}[(\text{Confinement}, \mathbf{M}_D^0)]. \quad (4.6)$$

For $f = L, U, D$, the associated terminal ROC is unformed. We denote the corresponding formation failure by $\perp_{f, \text{form}}$ in the status ledger; $\perp_{f, \text{form}}$ is not a terminal quotient and has no cardinality.

The word “parent” in (4.3) matters. The two positive child results do not cover the unresolved complement of their PCR-5E parent, and the parent tag cannot be forgotten without changing the theorem domain. Likewise, the two confinement residuals in (4.5) and (4.6) are separately kind-indexed. A common coloured source certificate does not identify the up- and down-coloured terminal targets.

4.2 Three roots, four spectral sectors, one bearer

For every ready neutral child the PCR-6 spectral theorem gives

$$\text{SpecCore}_b(X) = \{S_1, S_2, S_3, S_H\}, \quad \text{rank}(S_i) = 1, \quad \text{rank}(S_H) = 3, \quad (4.7)$$

where the three light sectors are nondegenerate and the heavy eigenspace is quotiented by its lawful $U(3)_H$ action to one standing heavy sector. This is a $3 + 1$ decomposition of the spectral core inside one certified neutral bearer. It is not a decomposition of I_3 , and S_H is not a fourth root [3].

Lemma 4.2 (Root-spectrum-bearer separation). *On a ready neutral child,*

$$|I_3| = 3, \quad |\text{SpecCore}_b(X)| = 4, \quad |\text{TermCar}_b(X)| = 1,$$

and none of these three cardinalities can be substituted for another.

Proof. The first cardinality belongs to the premetric occurrence-root object and is transported to the three structural response slots by (4.1). The second belongs to the target-relative spectral quotient (4.7): three simple light projectors remain distinguished, while all heavy rank-one presentations descend to the single invariant projector S_H . The third belongs to the complete terminal continuation quotient, whose unique class requires the full light-heavy source-connected support, common boundary, active closure, and matching packet. The defining witnesses and quotient relations are therefore different at all three levels. No map in the frozen handoff identifies their objects or their counting problems. $\square$

4.3 The exact terminal theorem

Define the universally activated terminal locus at the frozen evidence bases by

$$\text{Act}_{\text{all}}^{\text{term}} := \left\{ (\tau_N, \tau_L, \tau_U, \tau_D) \left| \begin{array}{l} \tau_f \in \text{ActCert}_f^+ \quad (f \in \mathfrak{F}), \\ \text{all four packets share one certified physical target} \end{array} \right. \right\}.$$

This set is not a space of possible worlds. It is the domain of actually certified simultaneous terminal packets available to the frozen successor theory.

Theorem 4.3 (U05 exact terminal bifurcation). *At the frozen evidence bases:*

- (i) *each $b \in B_N^{\text{ready}}$ has exactly one certified terminal bearer, with the distinct $3 + 1$ spectral core of Lemma 4.2;*
- (ii) *the neutral parent complement retains $\text{NotReady}[\text{PCR5E}]$;*
- (iii) *the charged-lepton and the two coloured dispositions are exactly (4.4)–(4.6);*
- (iv) *the L, U, D terminal ROC objects are unformed and contribute no terminal cardinality; and*
- (v) $\text{Act}_{\text{all}}^{\text{term}} = \emptyset$.

*Consequently, at these frozen evidence bases and within the declared terminal language, the strongest cross-kind terminal statement is **TerminalPartial**, not universal terminal one-carrier and not terminal exact-three.*

Proof. Item (i) is the target-qualified PCR-6 bearer theorem together with its spectral and within-realization ROC theorems. Their hypotheses retain the child tag b , so they imply neither the parent statement nor a theorem for another matter kind [3]. The sealed parent and L, U, D values give (ii) and (iii). For the last three kinds the interface-formation field is missing; the terminal signature and its

quotient therefore cannot be formed. A cardinality expression for such a quotient would be ill typed, proving (iv).

An element of $\text{Act}_{\text{all}}^{\text{term}}$ would in particular contain certified positive L, U, D packets admitted through the N18 domain. The frozen handoff contains none: it contains the three readiness values and the corresponding formation failures. Hence the displayed set has no element, proving (v). This emptiness is evidence-relative and does not assert that a future enlarged evidence base cannot populate the domain. $\square$

Corollary 4.4 (No terminal completion by analogy). *Neither the four fibres' agreement over the common root and erased structural reduct nor the two positive neutral child theorems define a terminal object for L, U , or D . In particular, no terminal exact-three count can be obtained by copying the neutral result across kind tags.*

Proof. The common-root spans and erased-reduct comparisons preserve the objects and maps of (4.1); they do not construct full-target cross-kind equivalences. Terminal certificates and confinement-complete continuation signatures are not in the reduct. The neutral positive objects have different target tags and do not inhabit any of the missing N18 admission domains. Thus there is no lawful transport on which the proposed analogy could act. $\square$

Counterexample criterion 4.5 (Terminal-bifurcation falsifiers). *The U05 theorem is falsified by counting $\perp_{f,\text{form}}$; replacing NotReady by zero, Negative, or physical nonexistence; forgetting a neutral child tag; calling the PCR-6 $3+1$ spectral sectors four roots; calling the three light sectors three terminal carriers; treating the rank-three heavy sector as G_4 ; identifying the U and D confinement residuals; or admitting an L, U, D terminal packet without a new formation and role-occupancy certificate.*

Status theorem 4.6 (Terminal endpoint). *The neutral ready-child fibres are positive and one-bearer. The neutral parent complement and all three evaluated L, U, D fibres retain their exact readiness values, with unformed L, U, D terminal ROCs. This is an exact terminal bifurcation, not an incomplete universal theorem.*

5 One-Higgs structural factorisation and coupling naturality

The common Higgs statement needed here is a carrier statement, not a numerical scalar-sector solution. The frozen structural packet contains one bridge/fixation carrier

$$C_H := H_{\text{br/fix}}$$

through which the accepted rooted fermionic response incidences are formed. It does not identify C_H with an empirical vacuum expectation value, a Higgs pole, a potential coefficient, or a running parameter. Those are possible later metric or numerical representatives of an admitted physical response, not alternative definitions of the carrier.

5.1 The rooted incidence factorisation

For an object $X \in \mathcal{K}_f$, let $\mathcal{H}_{F,f}(C_H; X)$ be the groupoid of kind- f rooted structural incidences whose bridge leg is the fixed object C_H . An object of this groupoid records a root in $E_f(X)$, its rooted component, its response slot, and its incidence with C_H ; it contains no numerical coupling value.

Definition 5.1 (Fermionic Higgs-response map). *The accepted fermionic Higgs-response map is the family*

$$\beta_{F,f,X} : Q_f^Y(X) = \text{Disc}(A_f(X)) \longrightarrow \mathcal{H}_{F,f}(C_H; X)$$

which attaches to each response slot its frozen rooted incidence through C_H . Its composite with the canonical chain is

$$\text{Disc}(E_f(X)) \xrightarrow{\eta_{f,X}} \mathcal{S}_f(C_f; X) \xrightarrow{\chi_{f,X}} Q_f^Y(X) \xrightarrow{\beta_{F,f,X}} \mathcal{H}_{F,f}(C_H; X). \quad (5.1)$$

Writing $\beta_F := \coprod_{f \in \mathfrak{F}} \beta_{F,f}$ denotes the tagged coproduct of these four fibrewise maps. It does not erase the kind tags.

Because η_f has source $\text{Disc}(E_f)$, no preliminary map from $\text{Disc}(I_3)$ is silently substituted into (5.1). The common three-point object is present through the positioned fibre $E_f = I_3 \times \{r_f\}$; the accepted theorem chain remains literal.

Definition 5.2 (Bosonic electroweak Higgs descent). *Let $D_{H,EW}$ be the as-yet-authorized domain on which a licensed metric Higgs/electroweak realization exists. The symbol*

$$\beta_{H,EW} : D_{H,EW} \dashrightarrow \mathcal{P}_{H,EW}^{\text{metric}}$$

denotes the partial bosonic descent from the bridge/fixation carrier to an electroweak physical-response presentation. Its domain includes the required target, metric licence, vacuum/fixation, and response-authority fields.

The subscripts in β_F and $\beta_{H,EW}$ are substantive. The first is a total map on the accepted structural response slots. The second is a partial map to a licensed bosonic metric realization. Their domains, codomains, and proof obligations differ, so $\beta_F = \beta_{H,EW}$ is not a false equation but an ill-typed one.

5.2 One carrier and natural response

Every lawful arrow $u : X \rightarrow Y$ of $\mathcal{K}_f$ transports the complete rooted-response incidence and preserves the frozen carrier. Denote the induced transport by

$$\mathcal{H}_{F,f}(u) : \mathcal{H}_{F,f}(C_H; X) \longrightarrow \mathcal{H}_{F,f}(C_H; Y).$$

Theorem 5.3 (U06 one-Higgs structural factorisation). *For every $f \in \mathfrak{F}$, every accepted rooted fermionic response factors as (5.1) through the same bridge/fixation carrier C_H . If a purported second same-scope carrier C'_H supports the same complete incidence family without an additional standing witness, then $C'_H \cong C_H$ over every incidence leg and represents the same carrier. A nonisomorphic C'_H necessarily adds witness, role, or target load and therefore belongs to a different scope.*

Proof. The first statement is the accepted FCGS–I response-incidence handoff: η_f forms the three rooted components, χ_f gives their exact response-slot quotient, and $\beta_{F,f}$ records the common bridge/fixation incidence. The construction retains f and the root, so the common carrier does not identify distinct matter kinds.

For uniqueness, compare C'_H with C_H using the complete family of rooted incidence maps. If no standing-bearing field changes, the two presentations have identical source, target, response, continuation, and bridge witnesses. They are therefore isomorphic in the standing quotient and constitute one carrier in two presentations. If they are not isomorphic, some complete-signature field differs. That difference is a new standing witness, a new carrier role, or a changed target; in each case the candidate is no longer a second carrier inside the fixed comparison scope. Thus the theorem excludes a second same-scope Higgs carrier without excluding declared scalar structure in a larger domain. This is the structural one-Higgs claim used below. It is consistent with the independent carrier admission and no-carrier-to-value-amplification results of [6]. $\square$

Theorem 5.4 (U07 rooted coupling naturality). *On the maximal lawful same-target groupoid $\mathcal{K}_f$, $\beta_{F,f}$ is natural. For every $u : X \rightarrow Y$,*

$$\mathcal{H}_{F,f}(u) \circ \beta_{F,f,X} = \beta_{F,f,Y} \circ \mathbb{A}_f(u). \quad (5.2)$$

Consequently the full composite in (5.1) is natural on $\mathcal{K}_f$. This naturality neither populates an unformed terminal ROC nor extends across a cross-kind full-target Hom set.

Proof. A morphism of $\mathcal{K}_f$ preserves the complete target fingerprint, including the root, rooted quotient, response slot, Higgs incidence, target, and provenance rows. Transporting a response slot and then attaching its preserved C_H -incidence therefore gives the same incidence as attaching it first and transporting the complete incidence. This is (5.2). Composing that square with the naturality squares for η_f and χ_f proves naturality of (5.1). The proof uses only arrows of $\mathcal{K}_f$; it supplies no arrow where the full-target Hom set is empty and no terminal witness absent from the structural fingerprint. $\square$

5.3 The Yukawa representation boundary

A numerical Yukawa presentation requires more data than $\beta_{F,f}$: a licensed metric target, field bases, normalization, renormalization prescription, scale, and an authority record. If those data are available, write

$$Y_{f,\alpha} : \text{Dom}(Y_{f,\alpha}) \dashrightarrow \text{Mat}_{m \times n}(\mathbb{C})$$

for the resulting authority-indexed coordinate map. Its singular values, eigenvectors, phases, texture, and fitted entries are coordinates downstream of a rooted structural response. They are not objects of Q_f^Y merely because the same letter Y is conventional in both contexts.

Proposition 5.5 (No numerical Yukawa promotion). *No numerical Yukawa matrix, matrix dimension, rank, singular-value list, basis, texture, running coordinate, calibration, or fit can:*

- (a) *form or recount I_3 , E_f , $\mathcal{S}_f(C_f)$, or Q_f^Y ;*
- (b) *create a second same-scope Higgs carrier or identify β_F with $\beta_{H,EW}$;*
- (c) *activate an unformed terminal ROC; or*
- (d) *strengthen any value in (4.3)–(4.6).*

Proof. Every listed numerical object is defined only after the structural incidence and its additional metric and coordinate authority have been supplied. Its domain therefore presupposes the upstream objects in (a)–(c). Using such an object to create one of those presuppositions would reverse the formation order and would make the result depend on basis, normalization, scale, or fit. Those fields are not present in the structural quotient and cannot alter its connected components. Nor do they supply the missing complete terminal signatures recorded by the readiness values. Hence none of the four promotions is well typed. $\square$

Corollary 5.6 (No matrix-cardinality proof). *Three structural response slots are not three positive Yukawa eigenvalues, and a three-by-three matrix display is not a proof of structural exact-three. Both directions of inference are excluded.*

Counterexample criterion 5.7 (Higgs and coupling falsifiers). *The U06–U07 layer fails if a second same-scope Higgs carrier is inserted without a distinguishing standing witness; if β_F is drawn as a separate carrier rather than a response incidence through C_H ; if β_F and $\beta_{H,EW}$ are identified; if naturality is asserted outside the maximal lawful target domain; if C_H is replaced by a vev, pole mass, potential coefficient, or RG coordinate; or if a Yukawa matrix, rank, basis, texture, or fit is used to prove a root count or terminal activation.*

Status theorem 5.8 (One-Higgs endpoint). *The one-Higgs theorem is exact at structural carrier scope: all accepted fermionic rooted responses factor naturally through C_H . Bosonic electroweak descent remains a distinct partial map, and numerical Yukawa representatives retain their downstream status.*

6 Authority-indexed projection and the particle dictionary

The transition from a premetric response carrier to a familiar particle display is not an untyped relabelling. A physical projection has a target, a metric licence, a continuation domain, a degeneracy convention justified by the target, and an authority record. These data determine where the projection is defined and which distinctions it is allowed to display. They also prevent the displayed spectrum from becoming a second source of carrier identity.

6.1 Projection authority

Definition 6.1 (Projection-authority packet). *A projection-authority packet for kind f is a tuple*

$$\alpha = (T, \lambda_T, W_T, \sim_{\text{deg}, T}, B_T, \Gamma_T, \text{Prov}_T),$$

where T is the fixed target, λ_T is its metric licence, W_T is the standing and continuation witness, $\sim_{\text{deg}, T}$ is the lawful degeneracy relation, B_T is the boundary and calibration ledger, Γ_T is the permitted target-preserving transport groupoid, and Prov_T fixes source provenance. The packet is projection-complete only when every field is defined and the witness survives all transports in Γ_T .

The packet does not require every metric coordinate to be internally predicted. A mass or pole may be a boundary-calibrated coordinate while the projection itself is lawful. What is forbidden is allowing that coordinate to supply the missing carrier, target, or continuation witness.

For a projection-complete α , let $D_{f,\alpha}(X) \subseteq \mathcal{H}_{F,f}(C_H; X)$ be the full subgroupoid on responses for which W_T and λ_T apply. If the authority packet is incomplete, $D_{f,\alpha}(X)$ is not silently totalized.

Definition 6.2 (Authority-indexed partial metric projection). *An authority-indexed metric projection is a partial functor*

$$\Pi_{f,\alpha} : \mathcal{H}_{F,f}(C_H; X) \dashrightarrow \mathcal{P}_{f,\alpha}(T), \quad \text{Dom}(\Pi_{f,\alpha}) = D_{f,\alpha}(X),$$

together with a carrier-forgetful functor

$$U_{f,\alpha} : \mathcal{P}_{f,\alpha}(T) \longrightarrow \mathcal{H}_{F,f}(C_H; X)$$

such that

$$U_{f,\alpha} \circ \Pi_{f,\alpha} = \text{incl}_{D_{f,\alpha}}. \tag{6.1}$$

The objects of $\mathcal{P}_{f,\alpha}(T)$ may carry modes, poles, widths, masses, mixing coordinates, propagation data, and detector-facing records, but each retains its upstream kind, root, response, carrier, target, and authority tag.

Equation (6.1) is carrier preservation, not bijectivity. One upstream carrier may admit several metric modes, and several presentation vectors may lie in one lawful degeneracy orbit. In the PCR-6 neutral children, for example, the one bearer projects to three light nondegenerate sectors and one rank-three heavy degenerate sector. The projection displays that $3 + 1$ spectral architecture without creating four bearers or a fourth root.

Theorem 6.3 (U08 carrier-preserving partial descent). *For every projection-complete authority packet, an admitted physical projection is kind-, root-, response-, carrier-, target-, and degeneracy-preserving in the sense of (6.1). At the frozen evidence bases:*

- (i) *PCR-6 supplies populated projection domains for its two ready neutral child targets, with the degeneracy described in Lemma 4.2;*
- (ii) *the neutral parent complement remains readiness-valued; and*
- (iii) *this paper does not totalize a terminal physical projection for L, U, D , whose terminal ROCs remain unformed.*

The theorem is a partial-descent theorem and makes no physical nonexistence claim about an unpopulated domain.

Proof. The defining equality says that forgetting the metric presentation after projection returns the exact upstream response on which the projection was defined. Hence none of the retained upstream fields can change. The lawful degeneracy relation belongs to α , so the projection cannot split a degenerate orbit into independently standing carriers or merge distinctions that the target witness retains.

PCR-6 provides the required target, continuation, projection, and degeneracy fields on the two certified child scopes, proving (i). The remaining assertions follow from Theorem 4.3: the parent complement and the L, U, D branches do not supply the terminal objects needed for a total terminal projection in this paper. Leaving a partial functor undefined outside its certified domain is part of its exact type, not a negative theorem about future physical realization. The general licensing and non-generativity of metric description agree with [7]. $\square$

6.2 The factorised particle dictionary

A physical designation is the last field of a typed descent record. Let $\text{Name}_{f,\alpha}$ be the set of conventional designations licensed on the target T , and let $\ell_{f,\alpha} : \mathcal{P}_{f,\alpha}(T) \rightarrow \text{Disc}(\text{Name}_{f,\alpha})$ be its target-side labelling functor.

Definition 6.4 (Particle dictionary). *The particle dictionary is the partial composite record*

$$\begin{aligned} \text{Dict}_{f,\alpha} : \text{Disc}(E_f) &\dashrightarrow \text{Disc}(\text{Name}_{f,\alpha}), \\ e &\longmapsto \ell_{f,\alpha}(\Pi_{f,\alpha}(\beta_{F,f}(\chi_f(\eta_f(e))))), \end{aligned} \quad (6.2)$$

with domain consisting exactly of those e for which the response lies in $D_{f,\alpha}$. The full dictionary entry retains the tuple

$$(f, e, \eta_f(e), \chi_f \eta_f(e), \beta_{F,f} \chi_f \eta_f(e), \alpha, \Pi_{f,\alpha}(\cdots), \ell_{f,\alpha}(\cdots)).$$

Thus every designation factors through kind, root, rooted response, and a licensed physical projection.

The structural symbols G_1, G_2, G_3 , when used, designate the three unmarked Higgs-response slots only after a chosen marking of the S_3 -set. They are neither primitive family names nor an ordering by mass. Familiar particle names enter only through $\ell_{f,\alpha}$, after the kind and projection fields have already been fixed.

Theorem 6.5 (U10 dictionary factorisation). *Every particle designation admitted by FCGS-III factors as (6.2). Consequently:*

- (a) *a name or family index cannot form a root or response slot;*
- (b) *the same structural slot may receive different target-side presentations without changing its carrier identity;*

- (c) a missing projection authority leaves the designation map undefined, rather than authorizing a name-first completion; and
- (d) the up- and down-coloured dictionaries remain distinct even where their three-slot structural forms are isomorphic.

Proof. The first assertion is Definition 6.4. The labelling functor is the final arrow, so it cannot supply any object required to form its own domain; this proves (a). Carrier preservation (6.1) permits changes of licensed metric presentation while retaining the upstream response, proving (b). A partial composite has no value outside the domain of its partial factor, proving (c). Finally, the kind tag is the first retained coordinate of the full dictionary record and is preserved by every later factor. Hence an isomorphism of the erased three-slot structures does not identify the U and D dictionaries, proving (d). $\square$

6.3 Matrix boundary and fixed-authority global no-backflow

The formation order relevant to this paper is

$$\text{Source} \prec \text{Root} \prec \text{Response} \prec \text{TerminalCertificate} \prec \text{MetricProjection} \prec \text{Coordinate} \prec \text{Record}. \quad (6.3)$$

Some branches stop before terminal certification; some lawful structural uses do not require every later stage. In no branch is the order reversed.

Proposition 6.6 (U09 Yukawa and mass-representation boundary). *Yukawa matrices, mass matrices, singular values, eigenvalues, bases, phases, mixing matrices, poles, and widths are coordinate or record objects after an authorized response and projection. They may describe, calibrate, compare, or test that realization. They cannot form E_f , select a structural slot, prove its cardinality, create C_H , or activate a terminal ROC.*

Proof. Each listed datum requires a choice of physical target and at least one of a metric licence, field basis, normalization, scale, propagation law, or measurement protocol. These are downstream fields in (6.3). None occurs in the definitions of E_f , η_f , χ_f , or $\beta_{F,f}$. Equation (6.1) supplies a lawful forgetful map from an authorized presentation to its response, but no authority-producing map in the reverse direction. The forbidden conclusions therefore do not follow. $\square$

Theorem 6.7 (U15 fixed-authority global no-backflow). *Within the derivation system generated by the frozen handoffs, (6.3), and the licensed target-preserving rules, let d be any admitted downstream mass, matrix, singular value, pole, width, spectrum, fit, calibration, prediction, decay record, confinement record, or detector reconstruction. Then no derivation from d and that fixed support can reverse-authorize any earlier source, root, rooted response, carrier, Higgs bridge, terminal certificate, projection domain, or common physical apex in (6.3). A downstream datum may test or falsify an already typed realization, but it cannot supply the standing presupposed by its own definition.*

Proof. Every downstream object carries an authority ancestry obtained by repeatedly forgetting later presentation fields. Suppose d authorized a missing upstream object a . If the ancestry of d already contains a , the argument is circular: d presupposes what it purports to create. If the ancestry does not contain a , using d imports a source, target, witness, or comparison rule absent from the declared domain and hence changes scope. If no standing-bearing field changes, the proposed authorization is only a presentation change and cannot alter a quotient or status. These cases exhaust the reverse use. Therefore no backflow is licensed. Allowing d to contradict a predicted record is different: falsification removes or restricts a candidate inside an already formed comparison class and does not generate upstream standing. $\square$

Theorem 6.8 (U16 frozen-interface numerical boundary). *Structural closure, scale standing, parameter specification, calibration, fit compatibility, numerical readiness, and prediction readiness are distinct statuses. In the derivation system generated by the frozen handoffs and licensed target-preserving rules, in particular,*

$$\begin{aligned} \text{StructuralUniversal} &\not\Rightarrow \text{ParameterComplete}, \\ \text{FitCompatible} &\not\Rightarrow \text{Predicted}, \\ \text{Bounded} &\not\Rightarrow \text{ValueForced}. \end{aligned}$$

No numerical status strengthens a terminal disposition without a new target-relative terminal certificate.

Proof. Structural closure is proved in the premetric root-response categories. Every later numerical status requires additional authority fields and has its own quotient and witness conditions. A fit uses observation as calibration; prediction readiness requires the value or admissible class to be fixed without that target datum. A bound fixes an order or region rather than a unique point. The frozen rule ledger contains no coercion or constructor from the premise type of any displayed arrow to its conclusion type. Thus none of the displayed implications is derivable in that system. This is a frozen-interface nonderivability result, not an unrestricted semantic nonimplication claim over every possible extension. The no-backflow theorem prevents a later numerical success from supplying a missing terminal certificate. $\square$

Counterexample criterion 6.9 (Projection and dictionary falsifiers). *The U08–U10 or U15–U16 layer fails if a partial projection is silently totalized; if its target, licence, degeneracy, boundary, or provenance field is dropped; if one carrier with several modes is recounted as several carriers; if a heavy basis vector is promoted above its $U(3)_H$ orbit; if a particle name precedes the kind/root/response/projection factors; if a matrix, pole, fit, or detector record activates an upstream object; or if a calibrated or bounded value is reported as an internally forced prediction.*

Status theorem 6.10 (Projection endpoint). *The particle dictionary is exact as an authority-indexed partial descent. Its populated fibres are those for which the required target and projection authority have been certified. The construction preserves carrier identity, keeps numerical coordinates downstream, and closes fixed-authority global no-backflow without pretending that every physical fibre is already populated.*

7 The typed process fibration

Physical processes relate already typed carriers, sectors, boundaries, and records. They do not erase the distinctions among those roles. We therefore construct a universal process grammar whose fibres may be populated only by a target-relative witness. The universality lies in the available types and their composition law, not in a claim that every physical process fibre is already certified.

7.1 Process sorts and endpoint words

Let $\mathfrak{F}^\pm$ be the four kind tags equipped with particle/antiparticle orientation, and let $\text{Word}(\mathfrak{F}^\pm)$ be the set of finite endpoint words in that alphabet, including the empty word. Define the process-sort set

$$\text{PSort} := \{\text{Transition}, \text{Decay}, \text{Weak}, \text{Pair}, \text{Confinement}, \text{Record}\}.$$

The distinctions are retained even when one experimental event admits more than one description. A decay is not identified with its detector record; a confinement continuation is not identified with a colour-response slot; and a pair-creation incidence is not identified with the source that licenses its endpoint carriers.

Definition 7.1 (Typed process base). *The category $\mathcal{B}_{\text{Proc}}$ has objects*

$$q = (\tau, s, w_{\text{in}}, w_{\text{out}}),$$

where τ is a declared target, $s \in \text{PSort}$, and $w_{\text{in}}, w_{\text{out}} \in \text{Word}(\mathfrak{F}^{\pm})$. A morphism $v : q \rightarrow q'$ is a lawful target-preserving redescription or transport which preserves the process sort, endpoint orientations, source/boundary types, and the target authority ledger. Such morphisms are invertible; hence $\mathcal{B}_{\text{Proc}}$ is a groupoid.

A charged-current process may have different kind tags in its input and output words. That does not produce a cross-kind arrow in $\mathbf{K}_{\mathfrak{F}}^{\text{full}}$. A process incidence is neither invertible as a physical transition nor required to preserve the complete fermion-target fingerprint. The process base records its two endpoints precisely so that a relation between kinds cannot be mistaken for an identification of kinds.

Definition 7.2 (Process-witness fibre). *For $q = (\tau, s, w_{\text{in}}, w_{\text{out}})$, let $\text{Proc}(q)$ be the category of process-witness packets*

$$P = (\tau, s, w_{\text{in}}, w_{\text{out}}, \text{Src}, \text{Bd}, \text{Inc}, \text{Cont}, \text{Auth}, \text{Deg}, \text{Rec}, \text{Prov}, \text{Fail}).$$

Here source, boundary, incidence, continuation, authority, degeneracy, record, provenance, and failure fields are separately typed. Morphisms preserve all fields and commute with their structure maps. The fibre is allowed to be empty. A merely declared grammar object does not populate it.

Every $v : q \rightarrow q'$ transports a packet over q' back along the preserved target, endpoint, and authority fields. Denote this functor by $v^* : \text{Proc}(q') \rightarrow \text{Proc}(q)$. Complete-field preservation gives $(\text{id}_q)^* = \text{id}_{\text{Proc}(q)}$ and $(v'v)^* = v^*(v')^*$. Thus

$$\text{Proc}(-) : \mathcal{B}_{\text{Proc}}^{\text{op}} \longrightarrow \mathbf{Cat}$$

is a contravariant functor.

Definition 7.3 (Universal typed process fibration). *The total process category is the Grothendieck construction*

$$\mathcal{E}_{\text{Proc}} := \int_{q \in \mathcal{B}_{\text{Proc}}} \text{Proc}(q),$$

with projection

$$p_{\text{Proc}} : \mathcal{E}_{\text{Proc}} \longrightarrow \mathcal{B}_{\text{Proc}}. \quad (7.1)$$

A target fibre is populated only if it contains a packet whose authority, continuation, and failure fields are certified at that target.

Two process packets may be composed only when they have the same declared target, the output word and boundary of the first match the input word and boundary of the second, and the two authority ledgers admit a common continuation. Composition concatenates incidence and continuation data while retaining the ordered list of process sorts and all intermediate boundary records. Consequently composition does not collapse a transition followed by a record into a new primitive process sort.

Theorem 7.4 (Universal process grammar). *The projection p_{Proc} in (7.1) is a split Grothendieck fibration. Its cartesian lift of $v : q \rightarrow q'$ at $P' \in \text{Proc}(q')$ is the canonical arrow*

$$(q, v^* P') \longrightarrow (q', P').$$

The grammar is universal over the declared process sorts and endpoint words, while fibre population remains target-relative and may be empty.

Proof. The identity and composition laws for the pullback functors were established above. The standard Grothendieck construction therefore supplies, for every base arrow and object over its codomain, the displayed cartesian arrow. The chosen pullbacks make these lifts split. Every declared sort and endpoint profile is an object of the base, proving universality of the grammar. The construction nowhere asserts that $\text{Proc}(q)$ has an object; emptiness of a fibre is compatible with the fibration and records the absence of a certified physical packet rather than removing the type. $\square$

7.2 Target-relative population

The frozen sources populate only the fibres for which their complete target contracts apply.

Definition 7.5 (Population predicate). *For $q \in \mathcal{B}_{\text{Proc}}$, write $\text{Pop}(q)$ when $\text{Proc}(q)$ contains a packet with certified source, boundary, incidence, continuation, authority, provenance, and failure rows. Write $\text{Typed}(q)$ when only the grammar and its missing-field record are formed. Then*

$$\text{Pop}(q) \implies \text{Typed}(q),$$

but the converse is not asserted.

On each PCR-6 ready neutral child, the accepted process theorem supplies target-relative propagation/transition, charged-current weak boundary, heavy-state production or decay response where activated, light-heavy matching continuation, and reconstruction records. These witnesses retain their child tags and their distinction between the three light sectors, the one heavy standing sector, and the one terminal bearer. They do not populate every process sort for every neutral target [3].

For the evaluated L, U, D targets, FCGS-II supplies the process registry and its exact noncoercion audit, but no complete terminal population theorem. In particular,

$$\begin{aligned} \text{Pop}(\tau_U, \text{Confinement}, w_{\text{in}}, w_{\text{out}}) & \text{ is not certified at } \mathbf{M}_U^0, \\ \text{Pop}(\tau_D, \text{Confinement}, w_{\text{in}}, w_{\text{out}}) & \text{ is not certified at } \mathbf{M}_D^0. \end{aligned}$$

These are the process-level faces of the two exact confinement readiness values. They are not assertions that quark confinement is physically absent; they state that the confinement-complete terminal packets required here are not formed at the frozen bases [2]. Likewise, a pair-creation fibre remains only typed unless an accepted target contract supplies its complete witness packet.

7.3 Process noncoercion

Definition 7.6 (Process role map). *Every field of a process packet carries one of three disjoint role tags:*

$$\text{CarrierNode}, \quad \text{ProcessIncidence}, \quad \text{BoundaryRecord}.$$

Endpoint references have role CarrierNode only because their carriers were formed upstream. The process object itself has role ProcessIncidence, and its reconstructed output has role BoundaryRecord. No role-changing coercion is part of $\mathcal{E}_{\text{Proc}}$.

Lemma 7.7 (Relation objects are not terminal bearers). *A transition, decay edge, weak incidence, pair-creation relation, confinement continuation, or reconstruction map cannot by itself be an independently standing terminal bearer.*

Proof. Each listed object is typed by at least one endpoint, boundary, incidence, or record field and therefore presupposes the objects it relates or records. If its endpoint references are deleted, its source

or target signature becomes undefined; if the relation is deleted while the endpoint carrier remains, one continuation witness may be lost but the endpoint is not thereby redefined as the relation. The role tags are consequently deletion-distinct. Promoting the relation to bearer would erase this distinction and would require a new complete carrier signature and role-occupancy proof, neither of which is supplied by the process packet. $\square$

Theorem 7.8 (U11 typed process fibration and noncoercion). *The FCGS-III process layer satisfies all of the following.*

- (i) *transition, decay, weak interaction, pair creation, confinement, and record formation are distinct universal grammar sorts;*
- (ii) *lawful target-preserving redescription acts functorially through the split fibration (7.1);*
- (iii) *a cross-kind weak or transition incidence does not induce a cross-kind full-target equivalence;*
- (iv) *no process incidence or boundary record coerces to a carrier, root, terminal certificate, pole, lifetime, or common apex;*
- (v) *populated physical fibres remain target-relative; and*
- (vi) *the U/D confinement-complete terminal fibres remain unpopulated at $\mathbf{M}_U^0$ and $\mathbf{M}_D^0$, preserving (4.5) and (4.6).*

Thus U11 closes as a universal typed process grammar with an exact partial population theorem, not as universal process completion.

Proof. Items (i) and (ii) are Definitions 7.1 and 7.3 together with Theorem 7.4. A process with different input and output kind tags is represented by one base object containing both endpoint words. It is not an invertible morphism preserving either complete fermion-target fingerprint, proving (iii). Definition 7.6 and Lemma 7.7 prove the carrier part of (iv); Theorem 6.7 proves the corresponding prohibition for poles, lifetimes, and common-apex authority. The definition of Pop proves (v). Finally, the two frozen confinement residuals deny the complete population witnesses required by Pop while leaving the process types in the base, proving (vi). $\square$

Corollary 7.9 (Process evidence cannot repair terminal readiness). *An observed transition, decay candidate, weak event, pair-production event, confinement signature, or detector reconstruction may support, calibrate, or falsify a process model inside its authorized target. It cannot, without a new complete packet, replace the missing charged-lepton or confinement terminal fields or promote any value in (4.3)–(4.6).*

Counterexample criterion 7.10 (Process-fibration falsifiers). *The U11 layer fails if process sorts are collapsed; if a cross-kind weak edge is called a full-target isomorphism; if an empty fibre is silently populated; if a detector record is used as its own source authority; if a decay channel is promoted to a pole or proper lifetime without an unstable-state theorem; if pair creation is used to form the carrier it presupposes; if a colour response or hadronic record is substituted for the missing confinement-complete terminal packet; or if the neutral ready-child process records are universalized beyond their target tags.*

Status theorem 7.11 (Process endpoint). *The process grammar is universal and functorial, its role distinctions are noncoercive, and its populated fibres are exactly target-relative. The theorem preserves the neutral positive child packets and the unresolved charged and confinement branches simultaneously, which is the strongest branch-complete U11 result available at the frozen evidence bases.*

8 The simultaneous fermionic common apex

A common apex is not a row-wise juxtaposition of four terminal claims. It is one coinstantiated witness over one target state and one parameter point. In particular, a charged certificate at one global form cannot be combined with a coloured certificate at another, and a neutral child-scope certificate cannot lose its target tag merely because the four records are displayed together. This section makes that simultaneity requirement part of the witness type.

8.1 The witness type

Let

$$\mathcal{E}^0 = (\mathcal{E}_N^0, \mathfrak{E}_L^0, \mathfrak{E}_Q^0)$$

denote the frozen evidence package. The simultaneous-boundary theorem of FCGS–II fixes more than a generic parameter point [2]. Write State^0 for the set of target–parameter states supported by the package. For $f \in \mathfrak{F}$ and $\omega \in \text{State}^0$, let

$$\text{Spec}_{12,f}^0(\omega) := (\text{ChiralRep}_f(\omega), \text{Conjugation}_f(\omega), \text{Charge}_f(\omega), \text{Hypercharge}_f(\omega), \text{WeakInc}_f(\omega), \quad (8.1)$$

$$\text{HiggsParticle}_f(\omega), \text{HiggsLeft}_f(\omega), \text{ConventionBridge}_f(\omega), \text{ColorInc}_f(\omega), \quad (8.2)$$

$$\text{AnomLedgerInc}_f(\omega), \text{JointAnom}_f(\omega), \Gamma_\omega, \text{Boundary}_\omega, \text{ConfinementAuth}_\omega), \quad (8.3)$$

$$\text{Spec}_{12}^0(\omega) := (\text{Spec}_{12,f}^0(\omega))_{f \in \mathfrak{F}}. \quad (8.4)$$

This is a kind-indexed family of fourteen-field target specifications, with a typed `NotApplicable` value where a field does not apply. The subscript 12 names the U12/N12 obligation; it is not an arity. For $f \in \{L, U, D\}$, its component is the FCGS–II quantum-boundary specification. The neutral component is the corresponding certified child-scope boundary record; it is not an additional FCGS–II quantum-boundary theorem.

An element $\omega \in \text{State}^0$ fixes, in one tuple, the physical target, admissible parameter point, source branch, chiral convention, global form Γ_ω , boundary branch, and physical- and confinement-authority branches. Those shared coordinates agree across the family (8.4); charge, hypercharge, colour, chiral representation, and Higgs incidence remain kind-indexed values. The state also retains the certified charged mass-shape and metric-readout together with the selector and stratifier audit outputs at their established codomains. None of these coordinates is quotiented away in the common-apex problem.

Nine coordinates in this specification remain positive-completion residuals:

$$\begin{aligned} M_{12}^{\text{conv}} := \{ & \delta_{\text{repLedger}}^L, \delta_{\text{weakLedger}}^L, \delta_{\text{HiggsIncidence}}^L, \\ & \delta_{\text{colorRep}}^U, \delta_{\text{weakLedger}}^U, \delta_{\text{HiggsIncidence}}^U, \\ & \delta_{\text{colorRep}}^D, \delta_{\text{weakLedger}}^D, \delta_{\text{HiggsIncidence}}^D \} \subset M_{L,+}^0 \amalg M_{U,+}^0 \amalg M_{D,+}^0. \end{aligned} \quad (8.5)$$

Thus carrying a target coordinate is not a proof that its certificate fibre is inhabited. In particular, anomaly-ledger incidence is not joint anomaly cancellation; mass shape is not pole authority; and symbolic one-anchor readout is not a terminal metric certificate.

Let

$$\mathbf{A}_{12} := (\text{charge, anomaly, vacuum, Higgs, colour, source, interaction})$$

be the ordered resource family. For $\alpha \in \mathbf{A}_{12}$, let $\text{Raw}_{12,\alpha}(\omega)$ be the typed resource candidates at ω , and let $\text{Req}_{12,\alpha}^0(\omega)$ be the exact finite predicate list on that resource imported at its original target and authority from the simultaneous-boundary, Higgs, metric-licensing, and neutral-child source theorems [2, 6, 7, 3]. Define

$$\text{Cert}_{12,\alpha}(\omega) := \prod_{x \in \text{Raw}_{12,\alpha}(\omega)} \prod_{P \in \text{Req}_{12,\alpha}^0(\omega)} P(x). \quad (8.6)$$

Thus a certificate is a candidate together with every required proof; a displayed candidate alone does not inhabit the fibre. We use the readable names

$$\begin{aligned} & (\text{ChargeCert}, \text{AnomalyCert}, \text{VacuumCert}, \text{HiggsCert}, \text{ColorCert}, \text{SourceCert}, \text{InteractionCert})(\omega) \\ & := (\text{Cert}_{12,\alpha}(\omega))_{\alpha \in \mathbf{A}_{12}}. \end{aligned}$$

For a fixed ω , define the joint nonterminal certificate fibre

$$\begin{aligned} \text{JointFld}(\omega) &:= \text{ChargeCert}(\omega) \times \text{AnomalyCert}(\omega) \times \text{VacuumCert}(\omega) \times \text{HiggsCert}(\omega) \\ &\quad \times \text{ColorCert}(\omega) \times \text{SourceCert}(\omega) \times \text{InteractionCert}(\omega). \end{aligned} \quad (8.7)$$

Every factor in (8.7) is a possibly empty certificate fibre, not a displayed value. Thus, for example, a charge assignment without its joint anomaly and global-form incidence is not an element of the first two factors, and a propagator display without the relevant authority is not an interaction certificate. No residual in (8.5), and no certified base output retained by ω , is promoted to a term of one of these fibres without an explicit compatible constructor.

We use a uniform notation $\text{Term}_f^+(\omega; \mathcal{E}^0)$ for the actually certified positive terminal domain at ω . On the sealed handoffs this specializes to

$$\text{Term}_L^+ = \text{ActCert}_L^+, \quad \text{Term}_U^+ = \text{ActCert}_{Q,U}^+, \quad \text{Term}_D^+ = \text{ActCert}_{Q,D}^+,$$

with the evidence and target arguments retained. The neutral factor contains only target-tagged certified child-scope objects; it does not contain the unresolved parent complement.

For the coherence condition below, write $\text{NB}_{N,\Gamma}$ for projection of a certified neutral child record to its kind-indexed boundary specification. This notation introduces no new neutral quantum-boundary claim.

Let $\text{Req}_{12,\text{proc}}^0$ be the U11 process-compatibility predicate family and $\text{Req}_{12,\text{term}}^0$ the U05 terminal-assignment compatibility family, both with their full indices. The exact coherence relation family is

$$\mathfrak{R}_{12}^{\text{req}} := \left(\prod_{\alpha \in \mathbf{A}_{12}} \text{Req}_{12,\alpha}^0 \right) \text{II} \text{Req}_{12,\text{proc}}^0 \text{II} \text{Req}_{12,\text{term}}^0. \quad (8.8)$$

This family is fixed before U12. The later U13 question is whether the module carrying a proved predicate is deletion-essential; it does not define the predicates used here.

Definition 8.1 (Simultaneous common-apex witness). *For $\omega \in \text{State}^0$, let $\text{Coh}_\omega(j, \tau)$ be the proposition that*

- (i) $j \in \text{JointFld}(\omega)$ and $\tau = (\tau_f)_{f \in \mathfrak{F}}$ with $\tau_f \in \text{Term}_f^+(\omega; \mathcal{E}^0)$;
- (ii) all entries lie over exactly the same target-parameter state ω . Thus their target, parameter, source, global-form, boundary, and authority coordinates agree, while every kind tag and every kind-indexed field of $\text{Spec}_{12,f}^0(\omega)$ remains present;
- (iii) the terminal entries satisfy

$$\text{QB}_{f,\Gamma_\omega}(\tau_f) = \text{Spec}_{12,f}^0(\omega) \quad (f \in \{L, U, D\}), \quad \text{NB}_{N,\Gamma_\omega}(\tau_N) = \text{Spec}_{12,N}^0(\omega).$$

The residual coordinates in (8.5) are not used as certificate terms;

- (iv) the Higgs entries factor through the one locked $C_H = H_{\text{br/fix}}$, without identifying the distinct fermionic incidences; and
- (v) every predicate in $\mathfrak{R}_{12}^{\text{req}}$ is realized by the same witness.

The certified common-apex set is the dependent sum

$$\text{ApexCert}^+(\mathcal{E}^0) := \coprod_{\omega \in \text{State}^0} \left\{ (j, \tau) \in \text{JointFld}(\omega) \times \prod_{f \in \mathfrak{F}} \text{Term}_f^+(\omega; \mathcal{E}^0) : \text{Coh}_\omega(j, \tau) \right\}. \quad (8.9)$$

The dependent sum in (8.9) is essential. Replacing it by a product of marginal existence statements would discard the shared ω , while replacing the terminal product by four structural three-slot fibres would change the claim from terminal standing to structural incidence.

Lemma 8.2 (Necessary terminal-factor projections). *For every $f \in \mathfrak{F}$, Definition 8.1 supplies a canonical projection*

$$\pi_f^{\text{term}} : \text{ApexCert}^+(\mathcal{E}^0) \longrightarrow \prod_{\omega \in \text{State}^0} \text{Term}_f^+(\omega; \mathcal{E}^0).$$

Likewise, projection to any factor in (8.7) is necessary for a common-apex witness.

Proof. An element of the dependent sum is a tuple $(\omega, j, (\tau_g)_{g \in \mathfrak{F}}, c)$, where c proves the coherence predicate. Sending that tuple to (ω, τ_f) gives the first map. The resource projections are the ordinary projections of j . No choice, comparison of names, or reconstruction from a marginal value is involved. $\square$

8.2 The frozen common-apex disposition

The sealed FCGS-II terminal handoff proves, relative to the evaluated bases [2],

$$\prod_{\omega \in \text{State}^0} \text{Term}_L^+(\omega; \mathcal{E}^0) = \prod_{\omega \in \text{State}^0} \text{Term}_U^+(\omega; \mathcal{E}^0) = \prod_{\omega \in \text{State}^0} \text{Term}_D^+(\omega; \mathcal{E}^0) = \emptyset. \quad (8.10)$$

Let

$$\text{NoAdmittedFactor}_{L,U,D} := (e_L^0, e_U^0, e_D^0), \quad e_f^0 : \prod_{\omega \in \text{State}^0} \text{Term}_f^+(\omega; \mathcal{E}^0) = \emptyset, \quad f \in \{L, U, D\}, \quad (8.11)$$

be the three-proof record supplied by (8.10). It is a certificate about admitted positive domains, not a residual profile and not a physical nonexistence theorem. These equalities concern admitted positive certificate domains. They do not say that charged leptons or quarks are absent, and they do not turn the corresponding readiness values into negative physical theorems.

Theorem 8.3 (U12 certified common-apex obstruction). *At the frozen evidence package,*

$$\boxed{\text{ApexCert}^+(\mathcal{E}^0) = \emptyset.} \quad (8.12)$$

The exact conclusion is failure of certified simultaneous activation at $\mathcal{E}^0$. It is not a theorem that no physical common apex exists, and it does not exclude a richer successor target carrying newly certified factors.

Proof. Suppose $a \in \text{ApexCert}^+(\mathcal{E}^0)$. By Lemma 8.2, $\pi_L^{\text{term}}(a)$ is an element of the first empty terminal domain in (8.10). This is impossible. The same contradiction can independently be obtained from the U or D projection. Hence the certified witness set is empty. Notice that the argument does not require a decision about whether the neutral child certificates or any individual resource fibre are populated; one necessary empty factor already blocks the dependent product.

The physical-existence question has a different domain. Definition 8.1 quantifies only over witnesses constructible from the frozen evidence package and admitted by its certificate rules. No exhaustivity theorem identifies that set with the class of all physically possible successor realizations. Therefore emptiness of the former cannot be eliminated into nonexistence of the latter. $\square$

Corollary 8.4 (Exact U12 status). *The common-apex claim has the evidence-relative disposition*

$$\text{Disp}_{\text{apex}}(\mathcal{E}^0) = \text{NotReady}[(\text{CommonApex}, \text{NoAdmittedFactor}_{L,U,D})].$$

The tag records an activation obstruction. It is neither $\text{Negative}[\text{PhysicalApex}]$ nor a zero-cardinality terminal ROC.

Proof. Theorem 8.3 proves that no positive certificate can be admitted, while the frozen handoff supplies no exhaustive negative theorem for physical common-apex existence. By the terminal disposition discipline, the only sound value is therefore the displayed readiness value. $\square$

Counterexample criterion 8.5 (Common-apex falsifiers). *The U12 result fails if a proposed witness combines different parameter points, global forms, source branches, boundaries, authority branches, or quantum-boundary target specifications; if a residual target coordinate is used as an activated certificate; if it omits any factor of (8.7); if it substitutes a structural root or response slot for a positive terminal certificate; if it forgets a neutral child tag; if it treats NotReady as a negative occupant; or if (8.12) is advertised as physical impossibility rather than certified-witness emptiness at $\mathcal{E}^0$.*

Status theorem 8.6 (Common-apex endpoint). *One simultaneous common-apex witness would have to contain all seven joint certificate fibres and one positive terminal certificate for every fermion kind at one target–parameter state. The frozen charged and coloured positive domains are empty, so no such certified witness forms. The exact endpoint is an evidence-relative activation obstruction with a declared successor rule.*

9 Deletion-essential endpoint core and search-design annex

Minimality here has a precise interface meaning. It is not brevity, logical independence, or the assertion that every theorem used inside a displayed module belongs to a minimum axiom basis. The endpoint has a fixed family of typed claim keys and a declared module granularity. A scientific core is minimal when literal deletion of any displayed module prevents recovery of at least one already proved claim at the same target, evidence base, proof class, and source rules.

Definition 9.1 (Frozen U13 claim family and owned-module granularity). *Let $\mathfrak{C}_{III}^{13}$ be the finite family consisting of the U01–U12 theorem fields; the exact neutral-child and neutral-parent fields; the L, U, D target, residual, disposition, formation, and successor fields; and the U15–U16 authority-boundary fields. Every key includes its target, evidence basis, proof class, and requested conclusion.*

The coordinates displayed in (9.1)–(9.9) are owned-module coordinates. A target key may index several dependent records, but its payload is owned once. A residual, status, and formation-failure theorem are grouped when one is an argument of the others. Their prose descriptions specify the content that must travel with each module; U13 does not assert logical independence of every internal theorem field.

Definition 9.2 (Certified-value fibre). *For a typed claim key q , let $\text{CertVal}(q)$ denote its certified value fibre. If q is propositional, this is $\text{Proof}(q)$. If q names target data, a residual, or a successor condition, it is the dependent pair of the typed value and the certificate that the value has the target, evidence, proof-class, source, and authority coordinates recorded by q . Thus target-data extraction is not silently treated as proof of an untyped proposition.*

Definition 9.3 (Licensed extractor category). *Let Ext_{III}^0 have as objects the tagged subpackets of the scientific core defined below. Its morphisms are finite composites of:*

- (i) *projection of an owned theorem, target-key, status, or boundary field;*

- (ii) a target-preserving reindexing explicitly proved in U01–U16;
- (iii) application of a named U01–U16 constructor when its complete typed premise support remains in the subpacket; and
- (iv) transport along an equality preserving target, evidence, proof class, source, and authority.

The named-constructor alternatives and their complete premise supports are the ones exhaustively enumerated in the producer ledger $\mathfrak{L}_{13}^{\text{prod}}$ below. A morphism may not cite a deleted module, import a new source, reconstruct an omitted proof payload, change a claim key, or use a downstream datum to form an upstream proof. These are exactly the frozen-source and no-backflow restrictions established by U01 and U15.

Definition 9.4 (No-repair deletion at owned-module granularity). *Let*

$$\Sigma_J = \prod_{j \in J} \Sigma_j$$

be the ownership-reduced module signature over the fixed ambient target and evidence package, and let $P \in \Sigma_J$. Write $\partial_j : \Sigma_J \rightarrow \Sigma_{J \setminus \{j\}}$ for deletion of module j and every typed reference to its payload. The dependent residual, status, and formation fields below have already been merged, so each ∂_j is a well-typed forgetful map. The module is deletion-essential relative to $\mathfrak{C}_{III}^{13}$ if there are a key $q_j \in \mathfrak{C}_{III}^{13}$ and its certified extractor

$$\pi_{q_j} : \Sigma_J \longrightarrow \text{CertVal}(q_j)$$

for which no extractor

$$\tilde{\pi}_{q_j} : \Sigma_{J \setminus \{j\}} \longrightarrow \text{CertVal}(q_j) \quad \text{satisfies} \quad \pi_{q_j} = \tilde{\pi}_{q_j} \circ \partial_j$$

in $\mathbf{Ext}_{III}^0$. Introducing a new theorem, source, target, coercion, or replacement field is a repair and lies outside the deletion test. The packet is no-repair minimal at the displayed module granularity when every displayed module is deletion-essential.

This definition distinguishes interface minimality from logical independence. A successor theorem may prove a missing result again from a richer source, but that is a repair and therefore a new packet. It does not show that deletion preserved the frozen handoff.

9.1 The common packet

The common packet contains exactly the objects used by more than one kind:

$$\mathcal{P}_{\text{com}}^{\text{min}} := (\mathbf{F}_0, \mathbf{R}_3, \mathbf{K}_{\text{max}}, \mathbf{H}_1, \mathbf{P}_{\text{auth}}, \mathbf{X}_{\text{proc}}, \mathbf{A}_0, \mathbf{B}_{\text{nb}}, \mathbf{Z}_{\text{num}}). \quad (9.1)$$

The coordinates are the following mathematical records.

$\mathbf{F}_0$, frozen identity.

The exact theorem-bearing input identity, evidence-package identity, versioned handoff identity, frozen claim language, and U01 import rule of $\mathbb{H}_{III}^{\text{in}}$. Source fingerprints and file names are integrity metadata of the U18 handoff, not fields of this scientific module. Kind-specific target values and successor rules occur only in the kind packets below.

R₃, positioned structural universality.

For each $f \in \mathfrak{F}$, the fixed positioned fibre $E_f = I_3 \times \{r_f\}$, the exact chain

$$\text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} Q_f^Y = \text{Disc}(A_f),$$

its exact-three component theorem, and the explicit fixed-kind identification required before replacing E_f by I_3 .

K_{max}, lawful comparison.

The four maximal same-target groupoids, typed cross-kind noncomparability, maximal-domain naturality, the erasures e_f , and the effective codescent comparison κ over the common structural reduct.

H₁, one-Higgs incidence.

The unique same-scope carrier $C_H = H_{\text{br}/\text{fix}}$, the maps $\beta_{F,f}$, and their rooted naturality, retaining $\beta_F \neq \beta_{H,EW}$.

P_{auth}, partial projection and dictionary.

The authority-packet type, the exact populated and unpopulated projection domains, carrier-preserving forgetful maps, lawful degeneracy fields, and the factorized particle dictionaries.

X_{proc}, process fibration.

The typed coherent-transition, weak-interaction, decay, pair-creation, confinement/channel, and record fibres together with their noncoercion and composition laws.

A₀, common-apex disposition.

The witness type of Definition 8.1, the factor projections, Theorem 8.3, and its evidence-relative claim boundary.

B_{nb}, fixed-authority global no-backflow.

The authority order and Theorem 6.7.

Z_{num}, numerical ceiling.

The separation of structural closure, scale, parameter, calibration, fit, numerical readiness, and prediction readiness in Theorem 6.8.

9.2 Kind-specific and relation packets

The kind packets retain data that cannot be obtained from the common structural reduct. Their target and evidence tags are coordinates, not annotations.

For each kind, write $\Phi_f^0 := \Phi_f(X_f^0)$ for the evaluated full-target fingerprint at the frozen input, and define its ownership-reduced marked target key by

$$\Theta_f^0 := \text{pr}_{\text{target-key}} \Phi_f^0. \quad (9.2)$$

The projection retains the kind, target, source, occurrence, root, response, registry, charge, colour, weak, chirality, Higgs-incidence, confinement, boundary, and authority coordinates required to identify the claim. It excludes status, residual, provenance-hash, failure, and successor payloads. Later occurrences of the target key are typed indices, not copied proof fields.

For the neutral kind, set

$$\mathcal{P}_N^{\min} := (\Theta_N^0, N_{\text{child}}, N_{\text{parent}}, N_{\text{type}}, N_{\text{succ}}), \quad (9.3)$$

where $\mathbf{N}_{\text{child}}$ contains the two separately tagged records $b_{\text{acc}}, b_{\text{glob}}$, their positive terminal certificates, the one-bearer theorem, and their distinct $3 + 1$ spectral cores; $\mathbf{N}_{\text{parent}}$ is exactly **NotReady**[PCR5E] on the uncovered parent complement; $\mathbf{N}_{\text{type}}$ is the type-separation theorem proving that the parallel rank-three heavy sector is not a fourth generation; and $\mathbf{N}_{\text{succ}}$ retains the counterexample and successor-target conditions [3]. These proof payloads occur in their named modules and not in Θ_N^0 .

The inherited experimental coordinates form a separately typed annotation [4],

$$\mathbf{A}_N^{\text{search}} := ([28, 30] \text{ GeV}_{\text{scout}}, \delta_{\text{ATLAS}}^{\text{reweight}} = 0.2 \text{ mm}, \text{SourceCertifiedSearchDesign, NoPoleClaim, NoProperLifetimeClaim}). \quad (9.4)$$

This annotation retains the source-certified 28–30 GeV scouting band and the exact ATLAS discrete response-reweight-grid coordinate 0.2 mm. It is useful for experimental scouting. It is not deletion-essential to the FCGS–III scientific theorem: deleting it does not alter the heavy-sector type separation, terminal bifurcation, or FCGS-FINAL.

For the charged kind, first form the single dependent disposition module

$$\mathbf{D}_L^0 := (\mathbf{M}_L^0, \text{NotReady}[(\text{CL}, \mathbf{M}_L^0)], \perp_{L, \text{form}}). \quad (9.5)$$

Then set

$$\mathcal{P}_L^{\min} := (\Theta_L^0, \mathbf{D}_L^0, \text{Succ}_L). \quad (9.6)$$

The residual profile $\mathbf{M}_L^0$ is the sealed three-part formation, positive-completion, and exhaustive-negative profile. The readiness and unformed-ROC fields depend on it and therefore travel in the same owned module; they are not independent copies of the residual.

The one shared coloured structural theorem and its two marked uses form one relation module

$$\mathbf{Y}_{UD}^0 := (\mathbf{YM}_{\text{joint}}, \rho_U, \rho_D, \text{NoDup}_{UD}), \quad (9.7)$$

where ρ_U and ρ_D are target-use reindexing maps. They are not projections or copies of sector objects, and **NoDup**_{UD} records precisely that one joint certificate is credited at two distinct typed uses.

For $k \in \{U, D\}$, define the dependent disposition module

$$\mathbf{D}_k^0 := (\mathbf{M}_k^0, \text{NotReady}[(\text{Confinement}, \mathbf{M}_k^0)], \perp_{k, \text{form}}), \quad k \in \{U, D\}. \quad (9.8)$$

The separate marked packet is

$$\mathcal{P}_k^{\min} := (\Theta_k^0, \mathbf{D}_k^0, \text{Succ}_k). \quad (9.9)$$

The two packets remain distinct even when their residual schemas are isomorphic. Their charge, Higgs-incidence, target, and successor fields are not erased.

Define the scientific core and complete endpoint record by

$$\mathcal{P}_{III}^{\min} := \mathcal{P}_{\text{com}}^{\min} \times_{\text{tag}} \mathcal{P}_N^{\min} \times_{\text{tag}} \mathcal{P}_L^{\min} \times_{\text{tag}} \mathbf{Y}_{UD}^0 \times_{\text{tag}} \mathcal{P}_U^{\min} \times_{\text{tag}} \mathcal{P}_D^{\min}, \quad (9.10)$$

$$\mathcal{P}_{III}^{\text{full}} := \mathcal{P}_{III}^{\min} \times_{\text{ann}} \mathbf{A}_N^{\text{search}}. \quad (9.11)$$

9.3 Proof-payload ownership and deletion witnesses

The ownership ledger for $\mathfrak{C}_{III}^{13}$ is:

F_0 . U01 import identity and frozen claim-language key.

R_3 . U04 positioned exact-three key.

$K_{\max}$. U02–U04 comparison, maximal-naturality, and erased-codescent keys.

H_1 . U06–U07 one-Higgs factorisation and rooted-naturality keys.

P_{auth} . U08–U10 projection-authority and particle-dictionary keys.

X_{proc} . U11 process-fibration and process-noncoercion keys.

A_0 . U12 simultaneous-apex witness and disposition keys.

B_{nb} and Z_{num} .

U15 no-backflow and U16 numerical-boundary keys, respectively.

Θ_f^0 . The marked target key $q_{\text{target},f}$, for each $f \in \mathfrak{F}$.

N_{child} , N_{parent} , and N_{type} .

The two ready-child terminal keys, the parent-complement disposition key, and the heavy-sector/not- G_4 key, respectively.

N_{succ} . The neutral successor-condition key.

D_f^0 , $f \in \{L, U, D\}$.

The dependent residual, status, and unformed-ROC/no-count key for the marked kind.

Succ_f , $f \in \{L, U, D\}$.

The marked successor-condition key.

Y_{UD}^0 . The one-theorem/two-use coloured Yang–Mills relation key.

The following tables define the finite producer/premise ledger $\mathfrak{L}_{13}^{\text{prod}}$. A parameterized row denotes one row for each displayed kind. Its domain is exactly the twenty-four modules of $\mathcal{P}_{III}^{\text{min}}$; within $\mathbf{Ext}_{III}^0$, the middle column is the complete list of licensed production paths for the owned key, and the last column names a typed premise removed by module deletion.

Module	Sole licensed path to an owned key	Premise absent after deletion
F_0	U01 import projection	frozen input identity and claim-language key
R_3	U04 positioned exact-three constructor	E_f, η_f, χ_f and its component proof
$K_{\max}$	U02–U04 comparison constructors	complete fingerprint, erasure, or naturality path
H_1	U06–U07 Higgs constructors	$C_H, \beta_{F,f}$, and rooted factorisation path
P_{auth}	U08–U10 projection/dictionary constructors	authority packet and certified projection domain
X_{proc}	U11 process constructors	process sort, noncoercion, and composition certificate
A_0	U12 apex constructor	witness type, factor projection, or disposition proof
B_{nb}	U15 firewall constructor	authority order and no-backflow proof
Z_{num}	U16 boundary constructor	typed separation of numerical authority layers

Module	Sole licensed path to an owned key	Premise absent after deletion
$\Theta_f^0, f \in \mathfrak{F}$	certified target-key projection of Φ_f^0	the marked full-target key itself
N_{child}	U05 neutral-child projection	tagged child certificate and one-bearer proof
N_{parent}	U05 complement projection	parent-complement boundary and readiness proof
N_{type}	neutral heavy-sector type constructor	light/heavy type-separation proof
N_{succ}	certified neutral successor projection	counterexample and successor conditions
$D_f^0, f \in \{L, U, D\}$	U05 marked disposition projection	dependent residual/status/unformed-ROC triple
$\text{Succ}_f, f \in \{L, U, D\}$	certified marked successor projection	kind-indexed successor condition
Y_{UD}^0	joint Yang–Mills constructor with ρ_U, ρ_D	the single joint theorem and no-duplication path

Lemma 9.5 (Unique ownership of endpoint proof payloads). *Every claim key in $\mathfrak{C}_{III}^{13}$ has exactly one owning module in $\mathcal{P}_{III}^{\min}$, and every displayed module owns at least one such key. Other modules may be indexed by the same ambient target, but none contains a second copy of the owned proof payload or a licensed U01–U16 constructor for that payload from retained fields alone.*

Proof. The common-module rows follow the already proved theorem types: U01 owns the frozen import rule; U02–U04 own comparison and positioned structural universality; U06–U07 own same-scope Higgs factorisation; U08–U10 own projection and dictionary authority; U11 owns the process fibration; U12 owns the simultaneous-apex construction; and U15–U16 own the authority and numerical boundaries. None of those proof types is a field of a kind-specific disposition module.

Conversely, the marked target keys, neutral child and parent results, heavy-sector type theorem, L, U, D dependent dispositions, successor conditions, and joint coloured use are not fields of the common structural modules. Equation (9.2) removes status, residual, failure, provenance-hash, and successor payloads from every Θ_f^0 . Equations (9.5) and (9.8) store each residual together with its dependent readiness and formation fields exactly once. The relation module stores one Yang–Mills theorem with two reindexed uses, not two theorem objects. The domain of $\mathfrak{L}_{13}^{\text{prod}}$ is the module family in (9.10), and its middle column exhausts the generators admitted by Definition 9.3. The listed ranges are disjoint at fixed claim key. This gives the stated ownership. $\square$

Lemma 9.6 (Owned-extractor nonfactorisation). *For every displayed module M in $\mathcal{P}_{III}^{\min}$, choose an owned key q_M from the ledger and let*

$$\pi_M : \mathcal{P}_{III}^{\min} \longrightarrow \text{CertVal}(q_M)$$

be its certified extractor. There is no morphism

$$\tilde{\pi}_M : \partial_M \mathcal{P}_{III}^{\min} \longrightarrow \text{CertVal}(q_M)$$

in $\mathbf{Ext}_{III}^0$ satisfying $\pi_M = \tilde{\pi}_M \circ \partial_M$.

Proof. Assume such a factorisation and inspect the final generator of $\tilde{\pi}_M$. It cannot be projection of an owned field, because Lemma 9.5 places the only payload in the deleted module. A lawful reindexing or equality transport requires an existing proof with the same target and proof-class codomain, which the same lemma excludes from the retained modules. A named U01–U16 constructor producing q_M would require the premise shown in the last column of $\mathfrak{L}_{13}^{\text{prod}}$, which deletion removes. The middle column exhausts the licensed producers by definition of $\mathbf{Ext}_{III}^0$. Finally, a nontrivial composite reduces to one of these cases at its last generator. The alternatives would import a new source, change the claim key, reconstruct the deleted payload by an unlicensed rule, or reverse authority, none of which is a morphism of $\mathbf{Ext}_{III}^0$. This contradiction proves nonfactorisation. $\square$

Theorem 9.7 (U13 ownership-exact no-repair minimality). *The scientific core $\mathcal{P}_{III}^{\min}$ is no-repair minimal relative to $\mathfrak{C}_{III}^{13}$ at the owned-module granularity of Definition 9.1. This is an interface-minimality theorem, not a claim that the internal premises of an owned module form a minimum axiom basis.*

The complete endpoint record $\mathcal{P}_{III}^{\text{full}}$ is not asserted to be minimal. Deleting A_N^{search} preserves every claim in $\mathfrak{C}_{III}^{13}$, whereas deleting N_{type} removes the heavy-sector/not- G_4 theorem.

Proof. Every displayed module owns a ledger key, and Lemma 9.6 proves that its selected extractor cannot factor through deletion. Hence every module of the scientific core is deletion-essential. The search annotation owns no key in $\mathfrak{C}_{III}^{13}$, so its deletion leaves all scientific extractors unchanged. $\square$

Corollary 9.8 (No lossy compression). *No summary retaining only counts, familiar particle names, an unqualified common-apex label, or unindexed dispositions preserves $\mathcal{P}_{III}^{\min}$. A lossless table or equivalent encoding of the same owned typed record is permitted.*

Counterexample criterion 9.9 (Minimal-packet falsifiers). *The U13 theorem fails if a payload is duplicated across modules; if minimality is asserted merely because a list is short; if an omitted coordinate is silently reconstructed; if a status label replaces its residual profile or proof class; if the two neutral child scopes are merged; if the neutral parent complement is omitted; if the joint coloured theorem is duplicated at U and D ; if an unformed ROC is stored as an empty set; or if target and successor keys are treated as optional prose. It also fails if the search-design annex is called deletion-essential or if deleting that annex is confused with deleting the heavy-sector type theorem.*

Status theorem 9.10 (Minimal scientific core and complete endpoint record). *The minimal scientific core is the ownership-reduced tagged product (9.10); the complete endpoint record is (9.11). Common coordinates are shared without erasing kind-specific fields, kind and relation coordinates are retained without proof-payload duplication, and the search-design annotation remains explicitly outside the minimality claim.*

10 Exact status bifurcation and the fermionic capstone

The final theorem is not obtained by selecting the most affirmative branch and suppressing the others. It is obtained by assigning each frozen claim key its exact proof-class and target-indexed status, then preserving the full typed family. This makes structural universality and terminal partiality compatible without allowing either statement to change the other's scope.

10.1 The six-way status space

Definition 10.1 (Six-way claim-status codomain). *For q in the ten-key aggregate family $\mathfrak{Q}_{14}^0$ defined below, let $\text{Payload}_s(q)$ be the type of certified payloads for status tag*

$$s \in \text{Tag}_6 := \{\text{StructuralUniversal}, \text{TerminalPartial}, \text{Split}, \text{Negative}, \text{TargetDependent}, \text{NotReady}\}$$

at exactly the target, evidence, proof class, authority, and conclusion coordinates of q . Define

$$\text{Status}_6(q) := \coprod_{s \in \text{Tag}_6} (\{s\} \times \text{Payload}_s(q)). \quad (10.1)$$

We write, for example, $\text{StructuralUniversal}_q[x]$ for the tagged injection of $x \in \text{Payload}_{\text{StructuralUniversal}}(q)$, and analogously for the other five tags. Positive neutral child theorems occur inside the payload of the aggregate terminal-partial key; they are not additional domain keys of Status_6 . Thus the absence of a seventh top-level Positive tag is intentional.

The charged/coloured split is the target-and-treatment decomposition required by the FCGS–II handoff and the v2 status register. It is not an emptiness or nonexistence predicate. Define

$$\mathbf{B}_{L|UD}^0(\omega) := ((T_L(\Gamma_\omega), \mathfrak{E}_L^0) \mid (T_U(\Gamma_\omega), \mathfrak{E}_Q^0), (T_D(\Gamma_\omega), \mathfrak{E}_Q^0)), \quad (10.2)$$

$$\mathbf{B}_{L|UD}^0 := (\mathbf{B}_{L|UD}^0(\omega))_{\omega \in \text{State}^0}, \quad (10.3)$$

$$\text{JointTerm}_{L|UD}^+(\mathcal{E}^0) := \coprod_{\omega \in \text{State}^0} \left(\text{Term}_L^+(\omega; \mathcal{E}^0) \times \text{Term}_U^+(\omega; \mathcal{E}^0) \times \text{Term}_D^+(\omega; \mathcal{E}^0) \right). \quad (10.4)$$

Lemma 10.2 (Charged/coloured target split and admission obstruction). *At the frozen evidence package,*

$$\text{JointTerm}_{L|UD}^+(\mathcal{E}^0) = \emptyset, \quad \text{Disp}_{L|UD}(\mathcal{E}^0) = \text{Split}[\mathbf{B}_{L|UD}^0]_{\text{target}}.$$

The split records the separate charged and coloured targets and proof treatments needed to preserve the imported theorem strength. The displayed emptiness is the distinct admission-level obstruction for the frozen joint terminal product. Neither conclusion states that no future common successor target exists.

Proof. Projection of any member of $\text{JointTerm}_{L|UD}^+(\mathcal{E}^0)$ to its charged coordinate would give an element of the empty admitted domain in (8.10); hence the product is empty.

For the split row, the charged branch is indexed by $(T_L(\Gamma_\omega), \mathfrak{E}_L^0)$, whereas the two coloured uses are indexed by $T_U(\Gamma_\omega)$ and $T_D(\Gamma_\omega)$ over the joint coloured evidence package $\mathfrak{E}_Q^0$. Their terminal factorisations, residual ledgers, and successor conditions are distinct, and no lawful full-target identification erases those indices. Thus separate target treatment is required, which is exactly the v2 meaning of $\text{Split}[\mathbf{B}_{L|UD}^0]_{\text{target}}$. This argument is independent of the preceding emptiness proof. $\square$

The exact atlas domain is the ten-key family

$$\mathfrak{Q}_{14}^0 := \{q_{\text{struct}}, q_{\text{terminal}}, q_{\text{split}}, q_{\text{apex,adm}}, q_{\text{neutral scope}}, q_{\text{neutral parent}}, q_L, q_U, q_D, q_{\text{apex,phys}}\}. \quad (10.5)$$

Each symbol denotes the full target-, evidence-, proof-class-, authority-, and conclusion-indexed key named by its subscript. For every $q \in \mathfrak{Q}_{14}^0$, let

$$\pi_q^{14} : \mathcal{P}_{III}^{\min} \longrightarrow \text{Status}_6(q)$$

be the U14 assembly morphism whose leaves are the unique certified-value extractors in the U13 ownership ledger. Define the non-owning view

$$\text{view}_{14} := (\pi_q^{14})_{q \in \mathfrak{Q}_{14}^0}, \quad \text{Atlas}_{III}^0 := \text{view}_{14}(\mathcal{P}_{III}^{\min}).$$

The atlas therefore stores no second proof payload: the following ten statements are equality paths

from the view to the uniquely owned fields:

$$\text{Atlas}_{III}^0(q_{\text{struct}}) = \text{StructuralUniversal}_{q_{\text{struct}}}[(\mathfrak{F}, I_3, C_H)], \quad (10.6)$$

$$\text{Atlas}_{III}^0(q_{\text{terminal}}) = \text{TerminalPartial}_{q_{\text{terminal}}}[\text{Disp}^{\text{term}}], \quad (10.7)$$

$$\text{Atlas}_{III}^0(q_{\text{split}}) = \text{Split}_{q_{\text{split}}}[\mathbf{B}_{L|UD}^0]_{\text{target}}, \quad (10.8)$$

$$\text{Atlas}_{III}^0(q_{\text{apex,adm}}) = \text{Negative}_{q_{\text{apex,adm}}}[\text{ApexCert}^+(\mathcal{E}^0) = \emptyset]_{\text{adm}}, \quad (10.9)$$

$$\text{Atlas}_{III}^0(q_{\text{neutral scope}}) = \text{TargetDependent}_{q_{\text{neutral scope}}}[B_N^{\text{ready}} \mid \text{parent} \setminus B_N^{\text{ready}}], \quad (10.10)$$

$$\text{Atlas}_{III}^0(q_{\text{neutral parent}}) = \text{NotReady}_{q_{\text{neutral parent}}}[\text{PCR5E}], \quad (10.11)$$

$$\text{Atlas}_{III}^0(q_L) = \text{NotReady}_{q_L}[(\text{CL}, \mathbf{M}_L^0)], \quad (10.12)$$

$$\text{Atlas}_{III}^0(q_U) = \text{NotReady}_{q_U}[(\text{Confinement}, \mathbf{M}_U^0)], \quad (10.13)$$

$$\text{Atlas}_{III}^0(q_D) = \text{NotReady}_{q_D}[(\text{Confinement}, \mathbf{M}_D^0)], \quad (10.14)$$

$$\text{Atlas}_{III}^0(q_{\text{apex,phys}}) = \text{NotReady}_{q_{\text{apex,phys}}}[(\text{CommonApex}, \text{NoAdmittedFactor}_{L,U,D})]. \quad (10.15)$$

The subscript `adm` in (10.9) is load-bearing. The bracket in that row is the proved emptiness payload carried by the `Negative` tag; it is not the proposition being negated. The row is a negative theorem about certified-apex existence at a fixed evidence package. Equation (10.15) is the distinct physical-apex claim key, for which no exhaustive impossibility theorem is available. The former therefore does not license replacing the latter by a negative physical status.

Theorem 10.3 (U14 exact six-way bifurcation). *The assignment Atlas_{III}^0 is well typed and branch-complete. For each fixed claim key it records exactly one proof-class/status value, and the values (10.6)–(10.15) preserve every accepted positive child scope, exact admission-level negative theorem, target split, target dependence, and readiness obstruction. No status can be strengthened by erasing its claim key.*

Proof. Structural universality is Theorem 3.5; terminal partiality and the neutral target dependence are Theorem 4.3; Lemma 10.2 proves the split, while the three L, U, D residual values are inherited without promotion from the sealed terminal handoff. Theorem 8.3 proves the admission-level negative row, while Corollary 8.4 proves the distinct physical-apex readiness row. These theorems have different codomains, so both are retained without contradiction.

Definition 10.1 uses a tagged disjoint union. Hence, for a fixed claim key, a term inhabits the summand justified by its proof and no other summand by implicit coercion. Changing target, evidence base, or proof class changes the claim key rather than the value. The ten distinct claim keys in $\mathfrak{Q}_{14}^0$ are exhausted by the ten displayed rows. Finally, Theorem 9.7 shows that deleting a branch tag, residual, or successor field destroys a required endpoint coordinate. Thus the displayed assignment is exact and branch-complete. $\square$

10.2 The firewall and numerical ceiling

The fixed-authority global firewall has already been proved at the level of formation and authority ancestry. Its capstone consequence can be stated compactly.

Corollary 10.4 (U15–U16 capstone ceiling). *No mass, matrix, singular value, pole, width, spectrum, fit, calibration, prediction, decay record, confinement record, detector reconstruction, or HNL scouting coordinate can:*

- (i) form or recount I_3 , E_f , $\mathcal{S}_f(C_f)$, or Q_f^Y ;
- (ii) create C_f or a second same-scope C_H ;

(iii) activate a missing terminal ROC or change any disposition in (10.11)–(10.14);

(iv) populate $\text{ApexCert}^+(\mathcal{E}^0)$; or

(v) turn a calibrated, bounded, or search-design coordinate into an internally forced numerical prediction.

The allowed numerical ceiling retains the source-certified HNL scouting band 28–30 GeV and the exact ATLAS discrete response-reweight-grid coordinate 0.2 mm only as search-design data, not as a pole mass, proper lifetime, or fourth generation [4].

Proof. Items (i)–(iv) are direct instances of Theorem 6.7 along the authority order (6.3). Each datum listed in the statement is formed after the object it would purportedly authorize; using it in reverse is either circular or a target change. Item (v) is Theorem 6.8. The two HNL coordinates carry an experimental-search authority only, so they remain within the stated ceiling and do not inhabit any pole, lifetime, root, or terminal-certificate type. $\square$

10.3 Fixed-package derivability and claim strength

Let Δ_{III}^0 denote the exact theorem-bearing support frozen by U01: the accepted FCGS–I and FCGS–II handoffs, the two neutral ready-child theorem packets, and the identified scientific source theorems used by U01–U16 at their declared scopes. The combined ARC plan and source fingerprints are excluded from Δ_{III}^0 .

Let $\mathfrak{Q}_{III}^0$ be the finite claim-key family consisting of:

- (i) the exact U01–U16 conclusion keys;
- (ii) the separately indexed structural, comparison, Higgs, projection, dictionary, process, neutral-child, neutral-parent, L, U, D , admission-apex, physical-apex, residual, formation, disposition, and successor keys; and
- (iii) the explicit nonpromotion keys for universal terminal one-carrier, terminal exact-three, physical common-apex activation, numerical forcing, and a second same-scope Higgs carrier.

Every key retains its target, evidence package, proof class, source authority, and requested conclusion. Let $\mathcal{L}_{III}^0$ be the corresponding finite typed claim language.

The old-language image of the imported theorem support is the formula family

$$\Delta_{III,\mathcal{L}}^0 := \{\varphi \in \mathcal{L}_{III}^0 : \varphi \text{ is an exact premise imported by U01 from } \Delta_{III}^0\}. \quad (10.16)$$

This definition retains each premise at its original claim key. It neither turns source identity metadata into a premise nor replaces an imported theorem by a stronger restatement.

Definition 10.5 (Frozen derivability). *The relation $\text{Der}_0(\Gamma; \varphi)$, for $\Gamma \subseteq \mathcal{L}_{III}^0$ and $\varphi \in \mathcal{L}_{III}^0$, is the least relation generated by:*

- (i) the identity rule $\varphi \in \Gamma \Rightarrow \text{Der}_0(\Gamma; \varphi)$;
 - (ii) the named, typed U01–U16 constructors applied to their complete premise lists; and
 - (iii) target-preserving equality transport, substitution, and finite composition of the preceding rules.
- No generator changes a target, evidence, proof-class, source, or authority coordinate. Define

$$\text{Cn}_0(\Gamma) := \{\varphi \in \mathcal{L}_{III}^0 : \text{Der}_0(\Gamma; \varphi)\}. \quad (10.17)$$

The exact boundary universe for the present comparison is

$$\mathfrak{B}_{III}^0 := \{b_{\text{source}}, b_{\text{kind}}, b_{\text{erasure}}, b_{\text{struct/term}}, b_{\text{Higgs}}, b_{\text{projection}}, b_{\text{process}}, b_{\text{apex}}, b_{\text{ownership}}, b_{\text{atlas}}, b_{\text{no-backflow}}, b_{\text{numerical}}, b_{\text{successor}}\}. \quad (10.18)$$

These symbols denote, respectively, canonical-source immutability; full-target cross-kind noncomparability; explicit-erasure/codescent scope; structural/terminal noncoercion; one-Higgs same-scope scope; partial projection and dictionary scope; process noncoercion; admission/physical-apex separation; owned-module minimality; the exact status atlas; the authority firewall; the numerical ceiling; and the kind-indexed residual and successor conditions. Each is the full typed boundary family proved under that heading, not an abbreviated slogan. A boundary in a larger language belongs to a successor comparison.

Definition 10.6 (Admissible fixed-package endpoint). *An admissible fixed-package endpoint record R retains the exact atlas on $\mathfrak{Q}_{14}^0$, has theorem family*

$$\text{Th}_0(R) \subseteq \text{Cn}_0(\Delta_{III, \mathcal{L}}^0),$$

has proved boundary family

$$\text{Bd}_0(R) \subseteq \mathfrak{B}_{III}^0,$$

and leaves unchanged the target, evidence, proof-class, source, and authority coordinates of every claim key. A changed target, enlarged evidence package, new source, or new inference rule defines a successor record rather than a member of this class. Let $\mathfrak{R}_{III}^0$ be the class of these records.

Definition 10.7 (Fixed-package endpoint preorder). *For $R, R' \in \mathfrak{R}_{III}^0$, write*

$$R \preceq_0 R'$$

when their claim-key coordinates agree,

$$\text{Th}_0(R) \subseteq \text{Cn}_0(\text{Th}_0(R')), \quad \text{Bd}_0(R) \subseteq \text{Bd}_0(R').$$

Write $R \prec_0 R'$ when $R \preceq_0 R'$ and not $R' \preceq_0 R$.

Lemma 10.8 (Fixed-package preorder). *The relation $\preceq_0$ is reflexive and transitive.*

Proof. Reflexivity follows from the identity generator of Der_0 and reflexivity of set inclusion. If $R \preceq_0 R'$ and $R' \preceq_0 R''$, every theorem of R has a finite derivation from the theorems of R' , and each premise from R' has a finite derivation from R'' . Substitution and finite composition in Definition 10.5 combine them into a derivation from R'' . Equality of claim-key coordinates and inclusion of boundary families are transitive. $\square$

Lemma 10.9 (Elementary closure laws). *For formula families $\Gamma, \Lambda \subseteq \mathcal{L}_{III}^0$, $\Gamma \subseteq \text{Cn}_0(\Gamma)$; if $\Gamma \subseteq \Lambda$, then $\text{Cn}_0(\Gamma) \subseteq \text{Cn}_0(\Lambda)$; and $\text{Cn}_0(\text{Cn}_0(\Gamma)) = \text{Cn}_0(\Gamma)$.*

Proof. Extensivity is the identity rule. Monotonicity follows by induction on a finite Der_0 -derivation. For idempotence, substitute the finite Γ -derivation of each premise into the outer derivation and use finite composition. $\square$

Let $\text{Th}_{\text{own}}^0(\mathcal{P}_{III}^{\min}) \subseteq \mathcal{L}_{III}^0$ be the formula image of the unique certified-value extractors in $\mathfrak{L}_{13}^{\text{prod}}$. Let

$$q_{13}^{\min} \in \mathcal{L}_{III}^0, \quad g_{13}^{\min} \in \text{CertVal}(q_{13}^{\min})$$

denote, respectively, the exact U13 minimality claim key and the proof object established by Theorem 9.7. Finally, let $\text{Graph}_{\mathcal{L}}(\text{view}_{14})$ be the ten equality formulas (10.6)–(10.15), at their full claim keys. Define the formula-valued final generator family

$$\begin{aligned} \mathbf{G}_{\text{FINAL}}^0 &:= \Delta_{III,\mathcal{L}}^0 \cup \text{Th}_{\text{own}}^0(\mathcal{P}_{III}^{\min}) \cup \{q_{13}^{\min}\} \\ &\cup \text{Graph}_{\mathcal{L}}(\text{view}_{14}) \subseteq \mathcal{L}_{III}^0. \end{aligned} \quad (10.19)$$

The occurrence of $\Delta_{III,\mathcal{L}}^0$ is a typed reference image owned by $\mathbf{F}_0$, not a second stored copy of the imported proofs. The U15 firewall and U16 numerical theorem occur once in $\mathcal{P}_{III}^{\min}$, as $\mathbf{B}_{\text{nb}}$ and $\mathbf{Z}_{\text{num}}$; subsequent displays refer to those proof payloads.

Lemma 10.10 (FCGS–III fixed-package saturation). *Within the declared claim language,*

$$\text{Cn}_0(\Delta_{III,\mathcal{L}}^0) = \text{Cn}_0(\mathbf{G}_{\text{FINAL}}^0). \quad (10.20)$$

Moreover, every boundary in $\mathfrak{B}_{III}^0$ is proved by the corresponding uniquely owned proof object referenced by the final record.

Proof. The imported family $\Delta_{III,\mathcal{L}}^0$ is a literal subset of $\mathbf{G}_{\text{FINAL}}^0$ by (10.19). Lemma 10.9 therefore gives the left-to-right inclusion in (10.20).

Conversely, every member of $\text{Th}_{\text{own}}^0(\mathcal{P}_{III}^{\min})$ is derived from the imported family by its U01–U16 production path in $\mathfrak{L}_{13}^{\text{prod}}$; $q_{13}^{\min}$ is derived by U13; and the ten graph formulas are derived by U14. The imported-family summand is available by the identity rule. Hence

$$\mathbf{G}_{\text{FINAL}}^0 \subseteq \text{Cn}_0(\Delta_{III,\mathcal{L}}^0).$$

Monotonicity and idempotence from Lemma 10.9 give the reverse closure inclusion.

Finally, U01 proves the source boundary; U02–U04 the kind, erasure, and structural/terminal boundaries; U06–U07 the same-scope Higgs boundary; U08–U11 the projection and process boundaries; U12 and U14 the apex and atlas boundaries; U13 the ownership boundary; U15 the no-backflow boundary; U16 the numerical boundary; and the kind packets the residual and successor boundary. These cases exhaust $\mathfrak{B}_{III}^0$. $\square$

10.4 FCGS-FINAL

Define the final fermionic record

$$\mathbb{F}_{\text{FINAL}}^{2,0} := (\mathcal{P}_{III}^{\text{full}}, g_{13}^{\min}, \text{view}_{14}), \quad (10.21)$$

with theorem and boundary components

$$\text{Th}_0(\mathbb{F}_{\text{FINAL}}^{2,0}) := \text{Cn}_0(\mathbf{G}_{\text{FINAL}}^0), \quad (10.22)$$

$$\text{Bd}_0(\mathbb{F}_{\text{FINAL}}^{2,0}) := \mathfrak{B}_{III}^0. \quad (10.23)$$

The atlas value is exposed as $\text{view}_{14}(\mathcal{P}_{III}^{\min})$; the record stores the view map, not a second copy of its owned payloads. The search annex is a non-generative annotation and does not enlarge (10.22). The record is dependent and tagged; it does not identify structural and terminal scopes or duplicate the U15–U16 proof payloads already owned by the scientific core.

Theorem 10.11 (U17 FCGS-FINAL and fixed-package greatest endpoint). *For the frozen FCGS-III target, evidence package, claim language, and proof constructors, $\mathbb{F}_{\text{FINAL}}^{2,0}$ is admissible and is a greatest element of the endpoint preorder:*

$$\forall R \in \mathfrak{R}_{III}^0, \quad R \preceq_0 \mathbb{F}_{\text{FINAL}}^{2,0}.$$

Consequently there is no admissible R with $\mathbb{F}_{\text{FINAL}}^{2,0} \prec_0 R$. This greatest-element result is relative to $\Delta_{III,\mathcal{L}}^0$, $\mathfrak{Q}_{III}^0$, the U01–U16 constructors, and $\mathfrak{B}_{III}^0$; it is not a completeness theorem for a larger source package, a successor target, or all fermionic physics. The record proves all of the following.

(i) For every $f \in \mathfrak{F}$, with $E_f = I_3 \times \{r_f\}$, the canonical chain is

$$\text{Disc}(E_f) \xrightarrow{\eta_f} \mathcal{S}_f(C_f) \xrightarrow{\chi_f} Q_f^Y = \text{Disc}(A_f),$$

and has exactly three rooted structural components. A display with source $\text{Disc}(I_3)$ is licensed only after the fixed-kind identification $p_f : E_f \xrightarrow{\sim} I_3$ is explicitly invoked.

- (ii) The chain is natural on every maximal lawful same-target groupoid; the explicitly erased charts have effective structural codescent, while full-target cross-kind Hom sets remain empty.
- (iii) Every accepted rooted fermionic response factors naturally through one same-scope bridge/fixation carrier $C_H = H_{\text{br}/\text{fix}}$.
- (iv) Projection, particle naming, and process composition are exact on their certified partial domains and preserve kind, root, response, carrier, target, authority, degeneracy, and process-role fields.
- (v) The neutral ready children retain one terminal bearer each and their distinct $3 + 1$ spectral core; the neutral parent complement and the L, U, D branches retain exactly (10.11)–(10.14), with unformed L, U, D terminal ROCs and no terminal cardinality assigned to them.
- (vi) The certified simultaneous common-apex witness set is empty at $\mathcal{E}^0$, while physical common-apex existence remains readiness-valued rather than negatively decided.
- (vii) The exact six-way atlas, module-minimal scientific core, retained search-design annex, fixed-authority global no-backflow theorem, and numerical ceiling are part of the endpoint.

Under the deliberately coarse claim-class projection π_{class} , the full dependent record has the summary

$$\begin{aligned} \pi_{\text{class}}(\mathbb{F}_{\text{FINAL}}^{2,0}) = & \text{StructuralUniversal}[\mathbf{R}_3, \mathbf{K}_{\text{max}}, \mathbf{H}_1] \\ & \oplus_{\text{tag}} \text{TerminalPartial}[\text{Disp}^{\text{term}}] \\ & \oplus_{\text{tag}} \text{NotReady}[\text{CommonApex}; \mathcal{E}^0]. \end{aligned}$$

The projection suppresses the admission-level negative apex theorem, neutral target dependence, individual residual rows, and search annex; it is not a replacement for the full record. No stronger universal terminal, numerical, or common-apex claim is contained in the projected summary.

Proof. Clause (i) is the positioned-fibre form of the accepted structural chain and Theorem 3.5. The fixed-kind qualification prevents a root-normalised shorthand from changing the canonical source. Clause (ii) combines Theorems 3.2, 3.4, and 2.5. Clause (iii) is Theorems 5.3 and 5.4. Clause (iv) follows from Theorems 6.3, 6.5, and 7.8.

Theorem 4.3 proves clause (v), and Theorem 8.3 with Corollary 8.4 proves clause (vi). Clause (vii) is Theorems 10.3, 9.7, and Corollary 10.4. These results inhabit distinct fields of the dependent record (10.21); no proof step changes a target tag, fills an empty certificate domain, or uses downstream data to create an upstream object.

The displayed generators are derived from $\Delta_{III,\mathcal{L}}^0$, the atlas is branch-complete, and every member of $\mathfrak{B}_{III}^0$ is proved, so the final record is admissible. Now let $R \in \mathfrak{R}_{III}^0$. By admissibility and Lemma 10.10,

$$\text{Th}_0(R) \subseteq \text{Cn}_0(\Delta_{III,\mathcal{L}}^0) = \text{Cn}_0(\mathbb{G}_{\text{FINAL}}^0) = \text{Th}_0(\mathbb{F}_{\text{FINAL}}^{2,0}).$$

The identity generator of Der_0 therefore gives the required theorem inclusion into $\text{Cn}_0(\text{Th}_0(\mathbb{F}_{\text{FINAL}}^{2,0}))$. Likewise,

$$\text{Bd}_0(R) \subseteq \mathfrak{B}_{III}^0 = \text{Bd}_0(\mathbb{F}_{\text{FINAL}}^{2,0}).$$

Membership in $\mathfrak{R}_{III}^0$ fixes all claim-key coordinates. Definition 10.7 now gives $R \preceq_0 \mathbb{F}_{\text{FINAL}}^{2,0}$. $\square$

10.5 Immutable BCRS/CSP handoff

Definition 10.12 (Frozen FCGS-III successor packet). *Let*

$$\mathfrak{H}_{III}^{17} := \{q_{17}^{\text{handoff}}\}$$

be the singleton typed handoff-key family whose unique key names the exact U17 greatest-endpoint theorem. This export key is disjoint from the U01–U16 old-language family $\mathcal{L}_{III}^0$: it indexes the proof object passed to U18, but is not added to $\mathbb{G}_{\text{FINAL}}^0$, Cn_0 , or $\text{ClaimCeiling}_{III}^0$. Let

$$g_{17}^{\text{handoff}} \in \text{CertVal}(q_{17}^{\text{handoff}})$$

be the proof object established by Theorem 10.11. Let view_{pc} be the non-owning projection from the unique U01–U16 theorem fields of $\mathbb{F}_{\text{FINAL}}^{2,0}$, together with g_{17}^{handoff} , to their ordered proof-class coordinates. Define

$$\text{ProofClass}_{U01:U17} := \text{view}_{\text{pc}}(\mathbb{F}_{\text{FINAL}}^{2,0}, g_{17}^{\text{handoff}}).$$

This map records the theorem, classification, naturality, status, noncoercion, minimality, or fixed-package saturation class stated by the corresponding theorem and Appendix A.

Likewise let view_{st} project the marked target, status, residual, formation, successor, and U14-view paths already owned by $\mathbb{F}_{\text{FINAL}}^{2,0}$. Define

$$\text{StatusTarget}_{III} := \text{view}_{\text{st}}(\mathbb{F}_{\text{FINAL}}^{2,0}).$$

It is an ordered reference view on $\mathfrak{Q}_{III}^0$, including the ten-key atlas domain $\mathfrak{Q}_{14}^0$; it does not duplicate the underlying payloads.

The source identity field $\text{SourceFingerprint}_{III}$ is the ordered nineteen-row object/role/version/byte-count/SHA-256 registry in the companion file `provenance/SOURCE_PROVENANCE.csv`. Appendix D identifies the primary handoff, direct-use, and governance rows. The other registry rows are conditional, contextual, or mediated support with their licensed uses and forbidden promotions recorded explicitly. Registry membership is not premise membership: the combined plan is tagged governance-only, and contextual citations do not enter Δ_{III}^0 .

The claim ceiling is the old-language closure

$$\text{ClaimCeiling}_{III}^0 := \{\varphi \in \mathcal{L}_{III}^0 : \text{Der}_0(\text{Th}_0(\mathbb{F}_{\text{FINAL}}^{2,0}); \varphi)\}.$$

Finally, $\text{SuccessorRule}_{III}$ is the requirement that any successor proof of an old-language conclusion outside this ceiling identify at least one newly tagged support item: a source theorem, certificate, target field, or inference rule, recorded in the appropriate premise or rule coordinate.

The BCRS/CSP handoff is the dependent record

$$\begin{aligned} \mathbb{H}_{III \rightarrow BC}^{2,0} := & (\mathbb{F}_{\text{FINAL}}^{2,0}, \text{ProofClass}_{U01:U17}, \text{StatusTarget}_{III}, \\ & \text{SourceFingerprint}_{III}, \text{ClaimCeiling}_{III}^0, \text{SuccessorRule}_{III}). \end{aligned}$$

It permits downstream use of structural exact-three, one-Higgs fermionic incidence, typed projection/process domains, the exact terminal atlas, and the admission-level common-apex obstruction. Within $\mathcal{L}_{III}^0$, the closure contains no licensed promotion to universal terminal one-carrier, a terminal exact-three count, confinement closure, unstable-pole authority, numerical Yukawa or mass forcing, a second same-scope Higgs carrier, or a physical common apex. Interacting Standard Model QFT completion and dynamical-gravity completion lie outside $\mathcal{L}_{III}^0$ and are not exported by this record. The fingerprint establishes object identity and version only; it does not strengthen any scientific conclusion.

Definition 10.13 (Claim-preserving conservative successor embedding). *Let $\mathbb{H}_{\text{next}}$ be a successor theory. A morphism*

$$\iota_{III} : \mathbb{H}_{III \rightarrow BC}^{2.0} \longrightarrow \mathbb{H}_{\text{next}}$$

is a claim-preserving conservative embedding when:

- (i) *theorem identity, proof class, status, target/evidence key, source fingerprint, claim ceiling, and every boundary field are faithfully preserved;*
- (ii) *each imported proof retains its identical old-language conclusion, rather than a logically stronger restatement;*
- (iii) *for every $\varphi \in \mathcal{L}_{III}^0$,*

$$\text{Der}_{\text{next}}\left(\text{Th}_{\text{next}}(\iota_{III}(\mathbb{H}_{III \rightarrow BC}^{2.0})); \varphi\right) \implies \text{Der}_0(\text{Th}_0(\mathbb{F}_{\text{FINAL}}^{2.0}); \varphi);$$

and

- (iv) *every genuinely new source, target, certificate, or inference field enters a tagged complementary packet $\mathbb{N}_{\text{next}}$, so a successor proof records whether its premise support leaves $\text{im } \iota_{III}$.*

Theorem 10.14 (U18 immutable, claim-bounded BCRS/CSP handoff). *The record $\mathbb{H}_{III \rightarrow BC}^{2.0}$ is a well-typed export interface. Let a BCRS or CSP successor receive it through a claim-preserving conservative embedding ι_{III} . If $\varphi \in \mathcal{L}_{III}^0$ has a successor proof whose premise support lies entirely in $\text{im } \iota_{III}$, then*

$$\varphi \in \text{ClaimCeiling}_{III}^0.$$

Consequently any old-language conclusion $\varphi \in \mathcal{L}_{III}^0$ outside that ceiling must use at least one newly tagged support item—a premise or an inference rule—and must be proved as a successor result. It cannot be presented as a restatement of FCGS-III.

Proof. Theorems 10.11 and 10.3 supply the final record and its exact proof-class, status, target, and boundary fields. Appendix D supplies the exact predecessor identities, while the displayed closure defines the ceiling. Hence the tuple inhabits Definition 10.12.

Now assume the premise support of a successor proof of φ is contained in $\text{im } \iota_{III}$. Old-language conservativity in Definition 10.13 returns an FCGS-III derivation of φ . By definition of $\text{ClaimCeiling}_{III}^0$, the conclusion lies in the ceiling. If a successor conclusion lies outside it, the support cannot remain inside the imported image; the tagged-complement condition forces at least one premise from $\mathbb{N}_{\text{next}}$. Such a proof is a new successor theorem with new provenance, not a strengthened restatement of the frozen handoff. $\square$

Counterexample criterion 10.15 (Capstone and handoff falsifiers). *The U14–U18 endpoint fails if the admission-level negative row is advertised as physical impossibility; if a NotReady, split, or target-dependent row is hidden; if $\text{Disc}(I_3)$ silently replaces the canonical positioned source $\text{Disc}(E_f)$; if a deletion-essential core module is omitted; if a mass, matrix, pole, fit, record, or scouting coordinate*

breaches the firewall; if any restatement assigns a different claim key or strength to an established theorem; if a source hash is used as a scientific premise; or if a terminal, numerical, confinement, pole, interacting-QFT, or gravitational successor theorem absent from the frozen packet is presented as an imported FCGS-III restatement rather than a newly supported result.

10.6 Conclusion

The fermionic architecture is universal at the level where universality has actually been proved: four positioned kind fibres carry one exact three-root structural geometry, three rooted response components each, and one shared bridge/fixation carrier, with maximal lawful same-target naturality and effective codescent only after explicit structural erasure. The theorem does not identify the full targets or turn a response slot into a terminal bearer.

Terminal physics bifurcates. Two target-qualified neutral child scopes carry one bearer with a distinct $3 + 1$ spectral core. Their parent complement and the charged and coloured branches retain exact readiness obstructions; the three L, U, D terminal ROCs are unformed and uncounted. Because a common apex requires all four positive terminal factors together with one complete joint certificate at one target-parameter state, its certified witness set is empty at the frozen evidence package. That is an exact activation result, not a declaration that future physical realization is impossible.

FCGS-FINAL is therefore greatest in the declared fixed-package preorder because it retains the complete proved theorem family and every exact boundary branch. It closes structural universality, preserves the exact terminal bifurcation, proves the fixed-authority global firewall and numerical ceiling, and exports an immutable claim-bounded handoff. Any later strengthening must identify at least one newly tagged source theorem, certificate, target field, or inference rule and must be proved as a separate successor result. The target changes only when the new support requires it.

A Realization of the U01–U18 obligations

For reference, the following list gives the mathematical disposition of every FCGS-III obligation. “Closed” means that the theorem stated in this paper has been proved from an accepted predecessor theorem or from a source used at its declared scope. “Typed” means that the exact domain or readiness value is part of the result. A typed row is not a failed proof: it is the theorem which prevents an absent certificate from being silently replaced by an analogy.

- U01 Import finality—closed.** Canonical FCGS-I, sealed FCGS-II, and both neutral ready-child results are frozen without promotion.
- U02 Fermion-kind category—closed.** Four tagged fibres lie over one structural base; full-target arrows require the complete fingerprint.
- U03 Maximal lawful naturality—closed/typed.** The maps η_f and χ_f are natural on their maximal lawful same-target domains; unlicensed cross-kind arrows remain absent.
- U04 Universal exact-three—closed.** Universality holds exactly at rooted structural-slot scope.
- U05 Universal terminality—typed.** The result is `TerminalPartial`: neutral child closure is positive, while the neutral parent complement and L, U, D retain exact readiness values.
- U06 One Higgs bridge—closed.** Every accepted rooted response factors through $C_H = H_{\text{br/fix}}$, with no second same-scope Higgs carrier.

- U07 Coupling naturality—closed/typed.** Structural-response naturality is closed; numerical Yukawa activation stays downstream and partial.
- U08 Metric projection—typed.** Projection is partial and authority-indexed; only certified domains are populated.
- U09 Yukawa boundary—closed.** Matrices, bases, singular values, and spectra generate neither carrier identity nor root or terminal-carrier cardinality.
- U10 Particle dictionary—typed.** Particle designations factor through kind, root, response, and a licensed physical projection.
- U11 Process fibration—closed/typed.** Process sorts and noncoercion are universal; populated terminal fibres remain target-relative.
- U12 Common Standard-Model apex—typed.** The FCGS-II simultaneous-boundary condition and its empty admitted L, U, D positive domains yield no certified witness at the frozen bases [2]. This is an activation obstruction, not physical impossibility.
- U13 Minimal packets—closed/typed.** The scientific core is module-minimal; the useful HNL search annotation is retained outside the minimality claim.
- U14 Exact bifurcation—closed.** The exact ten-key atlas exhausts its declared domain; structural-universal, terminal-partial, target-and-treatment split, negative, target-dependent, and NotReady values share one status codomain.
- U15 Fixed-authority global no-backflow—closed.** At a fixed authority, masses, matrices, poles, fits, and records cannot reverse-authorize an upstream object.
- U16 Numerical boundary—closed.** Structural closure is separated from scale, parameter, fit, and prediction readiness.
- U17 FCGS capstone—closed.** FCGS-FINAL is greatest in the declared fixed-package endpoint preorder and retains every exact readiness branch.
- U18 BCRS/CSP handoff—closed/conditional.** The export is claim-bounded under a claim-preserving conservative successor embedding, with exact fingerprints and new-proof provenance retained.

B Frozen status and object atlas

The integration theorem uses four objects that must not be conflated:

Object	What is counted	What it does not count
I_3	Three independently formed neutral occurrence roots	Particles, masses, terminal bearers, or heavy sectors
$\pi_0 Q_f^Y$	Three rooted structural-response components for a fixed kind	Positive couplings or metric poles
$\text{SpecCore}_b(X)$	In each certified neutral child realization, three light sectors and one rank-three heavy sector	Generation roots or terminal carriers
$\text{TermCar}_b(X)$	One terminal neutral bearer in each certified ready child realization	A universal bearer theorem for all neutrino or fermion targets

At the frozen evidence bases the terminal disposition map is

$$\begin{aligned} \text{Disp}_N(b) &= \text{Positive}[|\text{TermCar}_b(X)| = 1] & (b \in \{b_{\text{acc}}, b_{\text{glob}}\}), \\ \text{Disp}_N(\text{parent} \setminus \{b_{\text{acc}}, b_{\text{glob}}\}) &= \text{NotReady}[\text{PCR5E}], \\ \text{Disp}_L &= \text{NotReady}[(\text{CL}, \mathbf{M}_L^0)], \\ \text{Disp}_U &= \text{NotReady}[(\text{Confinement}, \mathbf{M}_U^0)], \\ \text{Disp}_D &= \text{NotReady}[(\text{Confinement}, \mathbf{M}_D^0)]. \end{aligned}$$

The last three values carry unformed terminal ROC objects. They are not zeros, and no terminal cardinality is assigned to them.

C Noncoercion matrix

Particle name or family index.

It may mark a licensed physical projection. It may not form I_3 , a response slot, or a terminal class.

Yukawa matrix or singular value.

It may represent a response after a basis and metric licence. It may not prove a three-root count or carrier identity.

Mass eigenvalue or pole.

It may describe a realized mode on an authorized target. It may not supply terminality, source completeness, or a common apex.

Oscillation or transition record.

It may distinguish standing light sectors on its target. It may not turn the heavy sector into a fourth root.

Confinement record.

It may witness a licensed coloured continuation. It may not replace the missing confinement-complete terminal packet.

Detector reconstruction.

It may report a response under a declared protocol. It may not authorize an upstream source, carrier, or lifetime.

HNL scouting pair.

It may define the source-certified 28–30 GeV scouting band and the exact ATLAS discrete response-reweight-grid coordinate 0.2 mm. It may not assert a pole mass, proper lifetime, or G_4 .

D Data availability and source provenance

The proofs in the text carry the scientific argument. Exact source identity and integrity metadata are supplied with this article in `provenance/SOURCE_PROVENANCE.csv`. Its nineteen rows record, for every accepted, direct, conditional, or contextual source, a stable source identifier, exact title and version, artifact identifier, byte length, SHA-256 digest, basis membership, licensed use, forbidden promotion, and verification status.

The register distinguishes the current coordinated structural editions from the retained source records of the preceding edition. The primitive-generation source is also recorded explicitly.

The accepted predecessor theorems are FCGS–I primitive-standing reference edition [1] and FCGS–II primitive-standing reference edition [2]; the direct neutral sources are PCR–6 [3] and PCR–8 [4]; and the direct structural support used here includes the Higgs-bridge and metric-licensing theorems [6, 7]. The combined ARC plan is separately identified in the registry as governance only: it fixes the U01–U18 acceptance contract but supplies no scientific premise. The registry fingerprints are reproducibility checks, not premises or substitutes for any proof.

E Exact nonclaims

No theorem in this paper asserts a universal terminal carrier, a terminal exact-three count, a fourth generation, a unique Standard Model parameter point, confinement closure, a universal unstable-pole construction, a numerical Yukawa matrix, a mass ordering, an HNL pole mass or lifetime, or a complete interacting Standard Model plus dynamical-gravity apex. Nor does a readiness value assert the physical nonexistence of the object whose certification remains open.

References

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